Zolotarev problem case `1a`

C. Poussot-Vassal*[1], I-V. Gosea[2], and A.C. Antoulas[3]

[1]ONERA, DTIS, Université de Toulouse, France
(`charles.poussot-vassal@onera.fr`)
[2]Max Planck Institute, CSC Group, Magdeburg, Germany
(`gosea@mpi-magdeburg.mpg.de`)
[3]Department of Electrical and Computer Engineering, Rice University, Houston
(`aca@rice.edu`)

July 19, 2026

Abstract

Zolotarev

Contents

*Corresponding author: `charles.poussot-vassal@onera.fr`

1 Detailed examples exposition and results

2 Approximation objective $r = 3$

2.1 Numerator and denominator coefficients

	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00044$		AAA-sign-L1000, $\sigma_3 = 0.0005$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	-0.011	$6.5 \cdot 10^{-4}$	-70.0	12.0
z^2	2.6	0	3.7	$1.8 \cdot 10^{-16}$	2.6	$-2.2 \cdot 10^{-3}$	0.95	0.064
z	0	0	$-6.8 \cdot 10^{-15}$	$-1.2 \cdot 10^{-15}$	$9.9 \cdot 10^{-3}$	$1.0 \cdot 10^{-3}$	-150.0	27.0
1.0	0.65	0	1.5	$1.9 \cdot 10^{-16}$	0.65	$-6.1 \cdot 10^{-4}$	-0.36	0.082
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-4.2 \cdot 10^{-15}$	$-1.3 \cdot 10^{-15}$	$-6.6 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	-180.0	31.0
z	2.2	0	4.2	$4.4 \cdot 10^{-16}$	2.3	$-1.9 \cdot 10^{-3}$	-0.38	0.084
1.0	0	0	$-2.8 \cdot 10^{-15}$	$-2.0 \cdot 10^{-16}$	$7.3 \cdot 10^{-3}$	$2.0 \cdot 10^{-4}$	-44.0	7.6

Table 1: Case `1a` (`cpv macos`), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_3 = 0.00037$		LF, $\sigma_3 = 0.0021$		AAA-sign-L200, $\sigma_3 = 0.00037$		AAA-sign-L1000, $\sigma_3 = 0.00037$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^3	0	0	0	0	$-2.4 \cdot 10^{-11}$	$2.7 \cdot 10^{-13}$	$-9.5 \cdot 10^{-15}$	$5.6 \cdot 10^{-13}$
z^2	2.6	0	3.7	$-1.8 \cdot 10^{-16}$	2.6	$-5.8 \cdot 10^{-15}$	2.6	$-4.2 \cdot 10^{-15}$
z	0	0	$-5.1 \cdot 10^{-15}$	$6.8 \cdot 10^{-16}$	$-5.4 \cdot 10^{-11}$	$6.3 \cdot 10^{-13}$	$-2.1 \cdot 10^{-14}$	$1.2 \cdot 10^{-12}$
1.0	0.65	0	1.5	$-2.4 \cdot 10^{-16}$	0.65	$-9.7 \cdot 10^{-15}$	0.65	$-6.4 \cdot 10^{-15}$
z^3	1.0	0	1.0	0	1.0	0	1.0	0
z^2	0	0	$-3.2 \cdot 10^{-15}$	$3.0 \cdot 10^{-18}$	$-6.2 \cdot 10^{-11}$	$7.1 \cdot 10^{-13}$	$-2.3 \cdot 10^{-14}$	$1.4 \cdot 10^{-12}$
z	2.2	0	4.2	$-4.0 \cdot 10^{-16}$	2.2	$-9.9 \cdot 10^{-15}$	2.2	$-9.8 \cdot 10^{-15}$
1.0	0	0	$-1.8 \cdot 10^{-15}$	$4.3 \cdot 10^{-16}$	$-1.6 \cdot 10^{-11}$	$1.9 \cdot 10^{-13}$	$-7.1 \cdot 10^{-15}$	$3.6 \cdot 10^{-13}$

Table 2: Case `1a` (`cpv winos`), $r = 3$, Z4: numerator (first lines) and denominator (last lines) coefficients.

2.2 Poles

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	0	$6.58 \cdot 10^{-16} + 4.81 \cdot 10^{-17}\imath$	$-0.00323 - 9.0 \cdot 10^{-5}\imath$	$-0.00181 + 0.493\imath$
2.0	$1.5\imath$	$2.14 \cdot 10^{-15} - 2.06\imath$	$0.00557 + 1.5\imath$	$-0.00165 - 0.494\imath$
3.0	$-1.5\imath$	$1.49 \cdot 10^{-15} + 2.06\imath$	$0.00428 - 1.5\imath$	$180.0 - 31.3\imath$

Table 3: Case `1a` (`cpv macos`), $r = 3$, Z4: poles.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	0	$4.35 \cdot 10^{-16} - 1.02 \cdot 10^{-16}\imath$	$6.9 \cdot 10^{-12} - 8.47 \cdot 10^{-14}\imath$	$3.42 \cdot 10^{-15} - 1.59 \cdot 10^{-13}\imath$
2.0	$1.5\imath$	$1.25 \cdot 10^{-15} - 2.06\imath$	$2.76 \cdot 10^{-11} + 1.5\imath$	$1.22 \cdot 10^{-14} + 1.5\imath$
3.0	$-1.5\imath$	$1.47 \cdot 10^{-15} + 2.06\imath$	$2.76 \cdot 10^{-11} - 1.5\imath$	$7.73 \cdot 10^{-15} - 1.5\imath$

Table 4: Case `1a` (`cpv winos`), $r = 3$, Z4: poles.

2.3 Zeros

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00044$	AAA-sign-L1000, $\sigma_3 = 0.0005$
1.0	NaN	NaN	$-0.00241 + 0.5\imath$	$-0.00236 + 1.2 \cdot 10^{-4}\imath$
2.0	$0.5\imath$	$9.48 \cdot 10^{-16} - 0.637\imath$	$-0.00245 - 0.501\imath$	$0.00667 - 1.48\imath$
3.0	$-0.5\imath$	$8.89 \cdot 10^{-16} + 0.637\imath$	$237.0 + 13.9\imath$	$0.00871 + 1.48\imath$

Table 5: Case `1a` (`cpv macos`), $r = 3$, Z4: zeros.

	Opt., $\sigma_3 = 0.00037$	LF, $\sigma_3 = 0.0021$	AAA-sign-L200, $\sigma_3 = 0.00037$	AAA-sign-L1000, $\sigma_3 = 0.00037$
1.0	NaN	NaN	$9.21 \cdot 10^{-12} + 0.5\imath$	$4.98 \cdot 10^{-15} + 0.5\imath$
2.0	$0.5\imath$	$7.16 \cdot 10^{-16} + 0.637\imath$	$9.2 \cdot 10^{-12} - 0.5\imath$	$2.0 \cdot 10^{-15} - 0.5\imath$
3.0	$-0.5\imath$	$6.45 \cdot 10^{-16} - 0.637\imath$	$1.09 \cdot 10^{11} + 1.22 \cdot 10^{9}\imath$	$7.82 \cdot 10^{10} + 4.66 \cdot 10^{12}\imath$

Table 6: Case `1a` (`cpv winos`), $r = 3$, Z4: zeros.

2.4 Poles and zeros visualisation

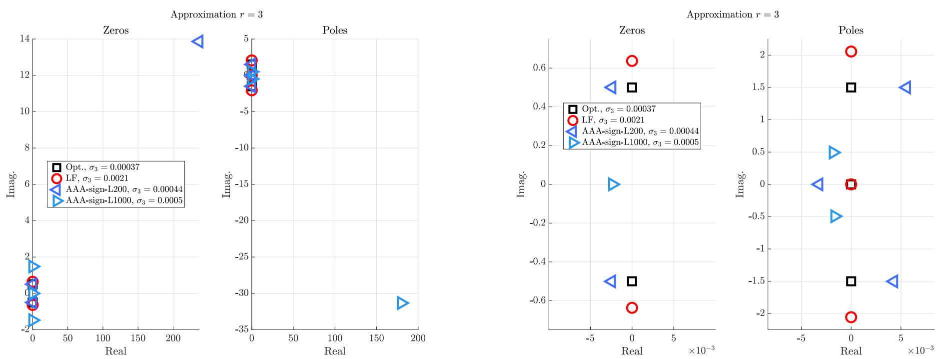

Figure 1: Case `'1a'`, $r = 3$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

2.5 Approximation visualization

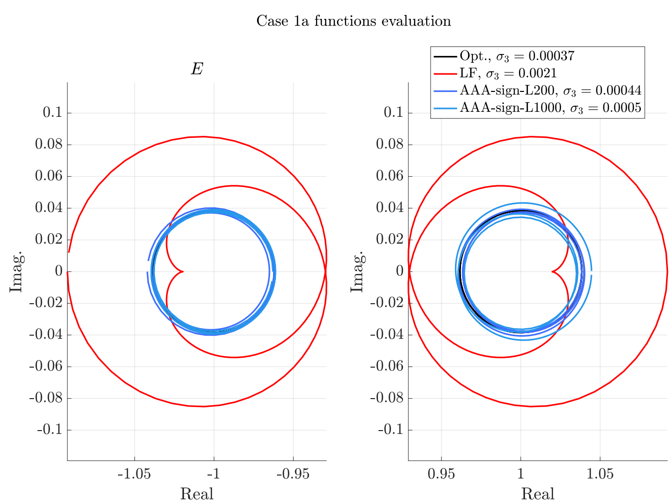

Figure 2: Case `'1a'`, $r = 3$. E and F evaluations.

3 Approximation objective $r = 4$

3.1 Numerator and denominator coefficients

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	-140.0	16.0	-150.0	7.3
z^3	3.5	0	3.7	$4.1 \cdot 10^{-16}$	4.7	-0.29	4.8	-0.22
z^2	0	0	$-4.6 \cdot 10^{-15}$	$3.8 \cdot 10^{-16}$	-650.0	70.0	-660.0	33.0
z	2.6	0	3.2	$2.0 \cdot 10^{-16}$	5.5	-0.52	5.6	-0.34
1.0	0	0	$-1.3 \cdot 10^{-15}$	$-1.6 \cdot 10^{-15}$	-80.0	8.8	-82.0	4.1
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-2.7 \cdot 10^{-15}$	$-8.4 \cdot 10^{-16}$	-500.0	54.0	-510.0	25.0
z^2	4.5	0	5.2	$8.2 \cdot 10^{-16}$	7.9	-0.7	7.9	-0.49
z	0	0	$-3.5 \cdot 10^{-15}$	$-1.1 \cdot 10^{-15}$	-370.0	41.0	-380.0	19.0
1.0	0.56	0	0.75	$1.0 \cdot 10^{-16}$	1.4	-0.13	1.4	-0.078

Table 7: Case 1a (cpv macos), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$		LF, $\sigma_4 = 6 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$		AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^4	0	0	0	0	$-5.4 \cdot 10^{-3}$	$-3.1 \cdot 10^{-3}$	$-4.8 \cdot 10^{-3}$	$-2.8 \cdot 10^{-3}$
z^3	3.5	0	3.7	$1.8 \cdot 10^{-16}$	3.5	$-3.7 \cdot 10^{-3}$	3.5	$-3.8 \cdot 10^{-3}$
z^2	0	0	$-2.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-16}$	-0.07	$-2.6 \cdot 10^{-3}$	-0.078	$-2.5 \cdot 10^{-3}$
z	2.6	0	3.2	$-5.5 \cdot 10^{-16}$	2.6	-0.011	2.6	-0.011
1.0	0	0	$-1.8 \cdot 10^{-15}$	$-7.7 \cdot 10^{-16}$	-0.015	$1.3 \cdot 10^{-3}$	-0.017	$1.4 \cdot 10^{-3}$
z^4	1.0	0	1.0	0	1.0	0	1.0	0
z^3	0	0	$-4.6 \cdot 10^{-16}$	$-1.2 \cdot 10^{-15}$	-0.036	$-6.9 \cdot 10^{-3}$	-0.038	$-6.5 \cdot 10^{-3}$
z^2	4.5	0	5.2	$2.7 \cdot 10^{-16}$	4.5	-0.011	4.5	-0.011
z	0	0	$-4.2 \cdot 10^{-15}$	$-4.2 \cdot 10^{-16}$	-0.054	$2.2 \cdot 10^{-3}$	-0.062	$2.5 \cdot 10^{-3}$
1.0	0.56	0	0.75	$-5.2 \cdot 10^{-16}$	0.57	$-3.3 \cdot 10^{-3}$	0.57	$-3.3 \cdot 10^{-3}$

Table 8: Case 1a (cpv winos), $r = 4$, Z4: numerator (first lines) and denominator (last lines) coefficients.

3.2 Poles

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09\imath$	$3.21 \cdot 10^{-16} - 0.384\imath$	$0.00377 + 6.65 \cdot 10^{-5}\imath$	$0.00375 - 2.07 \cdot 10^{-5}\imath$
2.0	$-1.94 \cdot 10^{-16} - 2.09\imath$	$3.09 \cdot 10^{-16} + 0.384\imath$	$0.00535 - 0.864\imath$	$0.00513 + 0.864\imath$
3.0	$-1.39 \cdot 10^{-17} + 0.359\imath$	$1.12 \cdot 10^{-15} - 2.25\imath$	$0.00516 + 0.864\imath$	$0.00532 - 0.864\imath$
4.0	$-1.39 \cdot 10^{-17} - 0.359\imath$	$9.09 \cdot 10^{-16} + 2.25\imath$	$499.0 - 54.4\imath$	$507.0 - 25.2\imath$

Table 9: Case 1a (cpv macos), $r = 4$, Z4: poles.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	$-1.94 \cdot 10^{-16} + 2.09\imath$	$5.75 \cdot 10^{-16} + 0.384\imath$	$0.00636 + 0.36\imath$	$0.00731 + 0.36\imath$
2.0	$-1.94 \cdot 10^{-16} - 2.09\imath$	$2.75 \cdot 10^{-16} - 0.384\imath$	$0.0052 - 0.361\imath$	$0.00612 - 0.361\imath$
3.0	$-1.39 \cdot 10^{-17} + 0.359\imath$	$-1.15 \cdot 10^{-16} - 2.25\imath$	$0.00935 - 2.1\imath$	$0.00961 - 2.09\imath$
4.0	$-1.39 \cdot 10^{-17} - 0.359\imath$	$-2.71 \cdot 10^{-16} + 2.25\imath$	$0.0147 + 2.1\imath$	$0.0149 + 2.1\imath$

Table 10: Case 1a (cpv winos), $r = 4$, Z4: poles.

3.3 Zeros

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 2.8 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 2.8 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00403 - 0.358\imath$	$0.00404 + 0.358\imath$
2.0	0	$4.1 \cdot 10^{-16} + 5.92 \cdot 10^{-16}\imath$	$0.00406 + 0.358\imath$	$0.00402 - 0.358\imath$
3.0	$0.866\imath$	$3.69 \cdot 10^{-16} + 0.929\imath$	$0.0128 - 2.08\imath$	$0.0127 - 2.09\imath$
4.0	$-0.866\imath$	$4.46 \cdot 10^{-16} - 0.929\imath$	$0.0117 + 2.09\imath$	$0.0116 + 2.09\imath$

Table 11: Case `1a` (`cpv macos`), $r = 4$, Z4: zeros.

	Opt., $\sigma_4 = 2.7 \cdot 10^{-05}$	LF, $\sigma_4 = 6 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_4 = 3.3 \cdot 10^{-05}$	AAA-sign-L1000, $\sigma_4 = 3.4 \cdot 10^{-05}$
1.0	NaN	NaN	$0.00562 - 4.73 \cdot 10^{-4}\imath$	$0.00656 - 5.2 \cdot 10^{-4}\imath$
2.0	0	$5.68 \cdot 10^{-16} + 2.27 \cdot 10^{-16}\imath$	$0.00528 - 0.87\imath$	$0.00611 - 0.869\imath$
3.0	$0.866\imath$	$3.38 \cdot 10^{-18} - 0.929\imath$	$0.00792 + 0.871\imath$	$0.00875 + 0.87\imath$
4.0	$-0.866\imath$	$1.73 \cdot 10^{-16} + 0.929\imath$	$486.0 - 282.0\imath$	$539.0 - 319.0\imath$

Table 12: Case `1a` (`cpv winos`), $r = 4$, Z4: zeros.

3.4 Poles and zeros visualisation

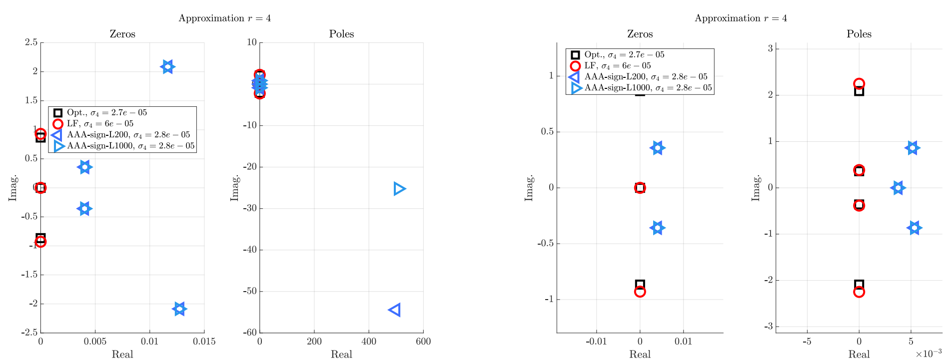

Figure 3: Case `'1a'`, $r = 4$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

3.5 Approximation visualization

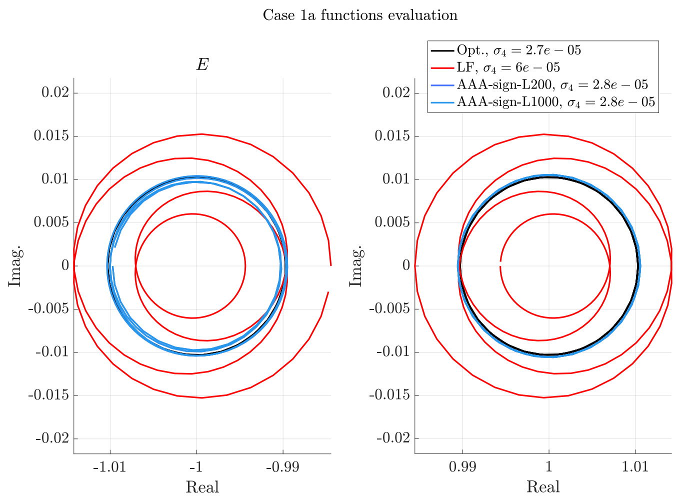

Figure 4: Case `'1a'`, $r = 4$. E and F evaluations.

4 Approximation objective $r = 5$

4.1 Numerator and denominator coefficients

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$-1.1 \cdot 10^{-11}$	$-4.8 \cdot 10^{-11}$	$2.1 \cdot 10^{-13}$	$4.5 \cdot 10^{-11}$
z^4	4.3	0	5.5	$-5.9 \cdot 10^{-15}$	4.3	$-6.4 \cdot 10^{-13}$	4.3	$1.2 \cdot 10^{-12}$
z^3	0	0	$-4.1 \cdot 10^{-15}$	$-3.4 \cdot 10^{-15}$	$-8.3 \cdot 10^{-11}$	$-3.6 \cdot 10^{-10}$	$1.5 \cdot 10^{-12}$	$3.4 \cdot 10^{-10}$
z^2	6.5	0	12.0	$-1.7 \cdot 10^{-14}$	6.5	$-2.9 \cdot 10^{-12}$	6.5	$5.2 \cdot 10^{-12}$
z	0	0	$-4.0 \cdot 10^{-15}$	$-9.0 \cdot 10^{-15}$	$-3.1 \cdot 10^{-11}$	$-1.4 \cdot 10^{-10}$	$5.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-10}$
1.0	0.49	0	1.1	$-1.3 \cdot 10^{-15}$	0.49	$-3.6 \cdot 10^{-13}$	0.49	$6.3 \cdot 10^{-13}$
z^5	1.0	0	1.0	0	1.0	0	1.0	0
z^4	0	0	$-3.6 \cdot 10^{-15}$	$-3.0 \cdot 10^{-15}$	$-4.8 \cdot 10^{-11}$	$-2.1 \cdot 10^{-10}$	$8.8 \cdot 10^{-13}$	$2.0 \cdot 10^{-10}$
z^3	7.5	0	11.0	$-1.6 \cdot 10^{-14}$	7.5	$-2.2 \cdot 10^{-12}$	7.5	$4.0 \cdot 10^{-12}$
z^2	0	0	$-4.4 \cdot 10^{-15}$	$-1.0 \cdot 10^{-14}$	$-7.2 \cdot 10^{-11}$	$-3.1 \cdot 10^{-10}$	$1.3 \cdot 10^{-12}$	$3.0 \cdot 10^{-10}$
z	2.8	0	5.8	$-7.1 \cdot 10^{-15}$	2.8	$-1.7 \cdot 10^{-12}$	2.8	$3.0 \cdot 10^{-12}$
1.0	0	0	$-9.8 \cdot 10^{-16}$	$-1.1 \cdot 10^{-15}$	$-5.4 \cdot 10^{-12}$	$-2.4 \cdot 10^{-11}$	$9.5 \cdot 10^{-14}$	$2.3 \cdot 10^{-11}$

Table 13: Case `1a` (`cpv macos`), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$		LF, $\sigma_5 = 1 \cdot 10^{-05}$		AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$		AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^5	0	0	0	0	$1.1 \cdot 10^{3}$	-31.0	$1.1 \cdot 10^{3}$	-31.0
z^4	4.3	0	5.5	$-3.7 \cdot 10^{-15}$	3.8	0.021	3.8	0.021
z^3	0	0	$-6.7 \cdot 10^{-15}$	$2.5 \cdot 10^{-15}$	$8.1 \cdot 10^{3}$	-240.0	$8.1 \cdot 10^{3}$	-230.0
z^2	6.5	0	12.0	$-1.4 \cdot 10^{-14}$	4.1	0.086	4.1	0.085
z	0	0	$-1.4 \cdot 10^{-14}$	$6.9 \cdot 10^{-17}$	$3.1 \cdot 10^{3}$	-89.0	$3.1 \cdot 10^{3}$	-87.0
1.0	0.49	0	1.1	$-1.9 \cdot 10^{-15}$	0.19	$9.6 \cdot 10^{-3}$	0.19	$9.5 \cdot 10^{-3}$
z^5	1.0	0	1.0	0	1.0	0	1.0	$-5.8 \cdot 10^{-17}$
z^4	0	0	$1.9 \cdot 10^{-16}$	$-1.4 \cdot 10^{-15}$	$4.7 \cdot 10^{3}$	-140.0	$4.7 \cdot 10^{3}$	-130.0
z^3	7.5	0	11.0	$-1.1 \cdot 10^{-14}$	5.7	0.071	5.7	0.07
z^2	0	0	$-1.7 \cdot 10^{-14}$	$3.9 \cdot 10^{-15}$	$7.1 \cdot 10^{3}$	-200.0	$7.1 \cdot 10^{3}$	-200.0
z	2.8	0	5.8	$-8.6 \cdot 10^{-15}$	1.4	0.047	1.4	0.046
1.0	0	0	$-3.3 \cdot 10^{-15}$	$5.1 \cdot 10^{-16}$	530.0	-15.0	530.0	-15.0

Table 14: Case `1a` (`cpv winos`), $r = 5$, Z4: numerator (first lines) and denominator (last lines) coefficients.

4.2 Poles

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$1.69 \cdot 10^{-16} + 1.84 \cdot 10^{-16}\imath$	$1.92 \cdot 10^{-12} + 8.44 \cdot 10^{-12}\imath$	$-3.41 \cdot 10^{-14} - 8.04 \cdot 10^{-12}\imath$
2.0	$3.89 \cdot 10^{-16} + 2.67\imath$	$2.45 \cdot 10^{-16} - 0.729\imath$	$2.84 \cdot 10^{-12} - 0.629\imath$	$-2.12 \cdot 10^{-13} + 0.629\imath$
3.0	$3.89 \cdot 10^{-16} - 2.67\imath$	$-1.57 \cdot 10^{-16} + 0.729\imath$	$3.03 \cdot 10^{-12} + 0.629\imath$	$1.08 \cdot 10^{-13} - 0.629\imath$
4.0	$-5.55 \cdot 10^{-17} + 0.629\imath$	$-1.18 \cdot 10^{-15} - 3.3\imath$	$1.98 \cdot 10^{-11} - 2.67\imath$	$-1.09 \cdot 10^{-12} + 2.67\imath$
5.0	$-5.55 \cdot 10^{-17} - 0.629\imath$	$4.72 \cdot 10^{-15} + 3.3\imath$	$2.05 \cdot 10^{-11} + 2.67\imath$	$3.47 \cdot 10^{-13} - 2.67\imath$

Table 15: Case `1a` (`cpv macos`), $r = 5$, Z4: poles.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	0	$5.62 \cdot 10^{-16} - 8.81 \cdot 10^{-17}\imath$	$-7.91 \cdot 10^{-5} + 0.281\imath$	$-7.91 \cdot 10^{-5} + 0.281\imath$
2.0	$3.89 \cdot 10^{-16} + 2.67\imath$	$7.72 \cdot 10^{-16} + 0.729\imath$	$-7.96 \cdot 10^{-5} - 0.281\imath$	$-7.95 \cdot 10^{-5} - 0.281\imath$
3.0	$3.89 \cdot 10^{-16} - 2.67\imath$	$2.77 \cdot 10^{-16} - 0.729\imath$	$-3.61 \cdot 10^{-4} + 1.19\imath$	$-3.61 \cdot 10^{-4} + 1.19\imath$
4.0	$-5.55 \cdot 10^{-17} + 0.629\imath$	$-2.67 \cdot 10^{-15} - 3.3\imath$	$-3.65 \cdot 10^{-4} - 1.19\imath$	$-3.65 \cdot 10^{-4} - 1.19\imath$
5.0	$-5.55 \cdot 10^{-17} - 0.629\imath$	$1.2 \cdot 10^{-15} + 3.3\imath$	$-4700.0 + 136.0\imath$	$-4700.0 + 134.0\imath$

Table 16: Case `1a` (`cpv winos`), $r = 5$, Z4: poles.

4.3 Zeros

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$2.08 \cdot 10^{-12} - 0.281\imath$	$-1.06 \cdot 10^{-13} + 0.281\imath$
2.0	$5.55 \cdot 10^{-17} + 1.19\imath$	$2.26 \cdot 10^{-16} - 0.32\imath$	$2.16 \cdot 10^{-12} + 0.281\imath$	$3.17 \cdot 10^{-14} - 0.281\imath$
3.0	$5.55 \cdot 10^{-17} - 1.19\imath$	$1.36 \cdot 10^{-16} + 0.32\imath$	$5.38 \cdot 10^{-12} - 1.19\imath$	$-4.15 \cdot 10^{-13} + 1.19\imath$
4.0	$6.94 \cdot 10^{-17} + 0.281\imath$	$5.92 \cdot 10^{-16} + 1.42\imath$	$5.74 \cdot 10^{-12} + 1.19\imath$	$2.15 \cdot 10^{-13} - 1.19\imath$
5.0	$6.94 \cdot 10^{-17} - 0.281\imath$	$-1.68 \cdot 10^{-16} - 1.42\imath$	$1.99 \cdot 10^{10} - 8.58 \cdot 10^{10}\imath$	$-4.5 \cdot 10^{8} + 9.58 \cdot 10^{10}\imath$

Table 17: Case `1a` (`cpv macos`), $r = 5$, Z4: zeros.

	Opt., $\sigma_5 = 1.9 \cdot 10^{-06}$	LF, $\sigma_5 = 1 \cdot 10^{-05}$	AAA-sign-L200, $\sigma_5 = 1.9 \cdot 10^{-06}$	AAA-sign-L1000, $\sigma_5 = 1.9 \cdot 10^{-06}$
1.0	NaN	NaN	$-6.3 \cdot 10^{-5} - 4.98 \cdot 10^{-6}\imath$	$-6.3 \cdot 10^{-5} - 4.9 \cdot 10^{-6}\imath$
2.0	$5.55 \cdot 10^{-17} + 1.19\imath$	$5.2 \cdot 10^{-16} - 0.32\imath$	$-1.45 \cdot 10^{-4} + 0.629\imath$	$-1.45 \cdot 10^{-4} + 0.629\imath$
3.0	$5.55 \cdot 10^{-17} - 1.19\imath$	$7.03 \cdot 10^{-16} + 0.32\imath$	$-1.46 \cdot 10^{-4} - 0.629\imath$	$-1.46 \cdot 10^{-4} - 0.629\imath$
4.0	$6.94 \cdot 10^{-17} + 0.281\imath$	$4.37 \cdot 10^{-16} + 1.42\imath$	$-0.00156 + 2.66\imath$	$-0.00156 + 2.66\imath$
5.0	$6.94 \cdot 10^{-17} - 0.281\imath$	$-4.16 \cdot 10^{-16} - 1.42\imath$	$-0.00158 - 2.67\imath$	$-0.00158 - 2.67\imath$

Table 18: Case `1a` (`cpv winos`), $r = 5$, Z4: zeros.

4.4 Poles and zeros visualisation

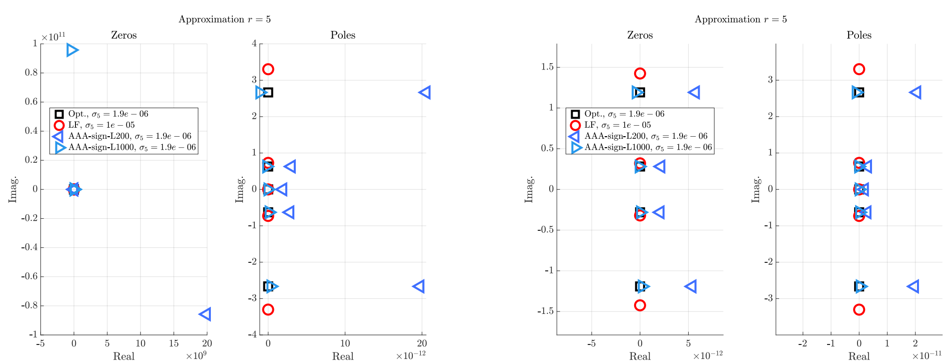

Figure 5: Case `'1a'`, $r = 5$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

4.5 Approximation visualization

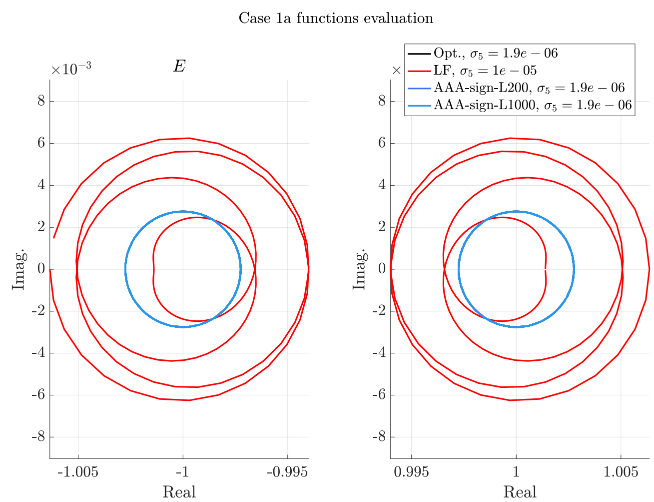

Figure 6: Case `'1a'`, $r = 5$. E and F evaluations.

5 Approximation objective $r = 6$

5.1 Numerator and denominator coefficients

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-1.2 \cdot 10^{-10}$	$-2.0 \cdot 10^{-10}$	$1.9 \cdot 10^{-12}$	$-8.0 \cdot 10^{-11}$
z^5	5.2	0	5.5	$-5.7 \cdot 10^{-15}$	5.2	$-6.3 \cdot 10^{-12}$	5.2	$-4.6 \cdot 10^{-12}$
z^4	0	0	$-2.3 \cdot 10^{-14}$	$-9.3 \cdot 10^{-15}$	$-1.4 \cdot 10^{-9}$	$-2.3 \cdot 10^{-9}$	$2.1 \cdot 10^{-11}$	$-9.2 \cdot 10^{-10}$
z^3	13.0	0	15.0	$-1.1 \cdot 10^{-14}$	13.0	$-4.7 \cdot 10^{-11}$	13.0	$-3.5 \cdot 10^{-11}$
z^2	0	0	$-4.2 \cdot 10^{-14}$	$-2.7 \cdot 10^{-14}$	$-1.1 \cdot 10^{-9}$	$-1.7 \cdot 10^{-9}$	$1.5 \cdot 10^{-11}$	$-7.0 \cdot 10^{-10}$
z	2.9	0	3.7	0	2.9	$-1.7 \cdot 10^{-11}$	2.9	$-1.3 \cdot 10^{-11}$
1.0	0	0	$-4.1 \cdot 10^{-15}$	$-3.2 \cdot 10^{-15}$	$-5.3 \cdot 10^{-11}$	$-8.7 \cdot 10^{-11}$	$7.8 \cdot 10^{-13}$	$-3.6 \cdot 10^{-11}$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$-6.9 \cdot 10^{-15}$	$-2.9 \cdot 10^{-15}$	$-6.5 \cdot 10^{-10}$	$-1.0 \cdot 10^{-9}$	$9.8 \cdot 10^{-12}$	$-4.2 \cdot 10^{-10}$
z^4	11.0	0	12.0	$-1.5 \cdot 10^{-14}$	11.0	$-2.7 \cdot 10^{-11}$	11.0	$-2.0 \cdot 10^{-11}$
z^3	0	0	$-4.3 \cdot 10^{-14}$	$-2.8 \cdot 10^{-14}$	$-1.6 \cdot 10^{-9}$	$-2.6 \cdot 10^{-9}$	$2.4 \cdot 10^{-11}$	$-1.1 \cdot 10^{-9}$
z^2	8.4	0	10.0	$-1.5 \cdot 10^{-15}$	8.4	$-4.0 \cdot 10^{-11}$	8.4	$-3.0 \cdot 10^{-11}$
z	0	0	$-2.0 \cdot 10^{-14}$	$-1.1 \cdot 10^{-14}$	$-3.6 \cdot 10^{-10}$	$-6.0 \cdot 10^{-10}$	$5.3 \cdot 10^{-12}$	$-2.5 \cdot 10^{-10}$
1.0	0.42	0	0.56	$-5.3 \cdot 10^{-16}$	0.42	$-3.0 \cdot 10^{-12}$	0.42	$-2.3 \cdot 10^{-12}$

Table 19: Case `1a` (`cpv macos`), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$		LF, $\sigma_6 = 3.2 \cdot 10^{-07}$		AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$		AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^6	0	0	0	0	$-4.4 \cdot 10^{8}$	$-2.2 \cdot 10^{9}$	$-6.7 \cdot 10^{7}$	$-6.4 \cdot 10^{9}$
z^5	5.2	0	5.5	$-4.0 \cdot 10^{-15}$	5.2	$-4.6 \cdot 10^{-4}$	5.2	$5.3 \cdot 10^{-4}$
z^4	0	0	$-2.4 \cdot 10^{-16}$	$1.1 \cdot 10^{-14}$	$-5.0 \cdot 10^{9}$	$-2.5 \cdot 10^{10}$	$-7.5 \cdot 10^{8}$	$-7.2 \cdot 10^{10}$
z^3	13.0	0	15.0	$-1.9 \cdot 10^{-14}$	13.0	$-3.2 \cdot 10^{-3}$	13.0	$3.8 \cdot 10^{-3}$
z^2	0	0	$-1.8 \cdot 10^{-15}$	$8.2 \cdot 10^{-15}$	$-3.7 \cdot 10^{9}$	$-1.9 \cdot 10^{10}$	$-5.6 \cdot 10^{8}$	$-5.4 \cdot 10^{10}$
z	2.9	0	3.7	$-5.4 \cdot 10^{-15}$	2.9	$-1.3 \cdot 10^{-3}$	2.9	$1.7 \cdot 10^{-3}$
1.0	0	0	$-4.3 \cdot 10^{-16}$	$3.7 \cdot 10^{-16}$	$-1.9 \cdot 10^{8}$	$-9.4 \cdot 10^{8}$	$-2.8 \cdot 10^{7}$	$-2.7 \cdot 10^{9}$
z^6	1.0	0	1.0	0	1.0	0	1.0	0
z^5	0	0	$1.8 \cdot 10^{-15}$	$-9.9 \cdot 10^{-17}$	$-2.3 \cdot 10^{9}$	$-1.2 \cdot 10^{10}$	$-3.5 \cdot 10^{8}$	$-3.3 \cdot 10^{10}$
z^4	11.0	0	12.0	$-1.1 \cdot 10^{-14}$	11.0	$-1.9 \cdot 10^{-3}$	11.0	$2.1 \cdot 10^{-3}$
z^3	0	0	$-4.6 \cdot 10^{-15}$	$1.6 \cdot 10^{-14}$	$-5.8 \cdot 10^{9}$	$-2.9 \cdot 10^{10}$	$-8.7 \cdot 10^{8}$	$-8.3 \cdot 10^{10}$
z^2	8.4	0	10.0	$-1.8 \cdot 10^{-14}$	8.5	$-2.8 \cdot 10^{-3}$	8.4	$3.6 \cdot 10^{-3}$
z	0	0	$7.3 \cdot 10^{-19}$	$3.8 \cdot 10^{-15}$	$-1.3 \cdot 10^{9}$	$-6.5 \cdot 10^{9}$	$-2.0 \cdot 10^{8}$	$-1.9 \cdot 10^{10}$
1.0	0.42	0	0.56	$1.4 \cdot 10^{-16}$	0.42	$-2.4 \cdot 10^{-4}$	0.42	$2.7 \cdot 10^{-4}$

Table 20: Case `1a` (`cpv winos`), $r = 6$, Z4: numerator (first lines) and denominator (last lines) coefficients.

5.2 Poles

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23\imath$	$8.54 \cdot 10^{-16} - 0.243\imath$	$1.9 \cdot 10^{-11} - 0.232\imath$	$-4.84 \cdot 10^{-13} - 0.232\imath$
2.0	$5.55 \cdot 10^{-17} - 3.23\imath$	$1.13 \cdot 10^{-15} + 0.243\imath$	$1.96 \cdot 10^{-11} + 0.232\imath$	$-7.24 \cdot 10^{-14} + 0.232\imath$
3.0	$-1.39 \cdot 10^{-16} + 0.866\imath$	$1.4 \cdot 10^{-15} - 0.908\imath$	$3.5 \cdot 10^{-11} - 0.866\imath$	$-1.3 \cdot 10^{-12} - 0.866\imath$
4.0	$-1.39 \cdot 10^{-16} - 0.866\imath$	$1.03 \cdot 10^{-16} + 0.908\imath$	$3.7 \cdot 10^{-11} + 0.866\imath$	$2.56 \cdot 10^{-13} + 0.866\imath$
5.0	$1.21 \cdot 10^{-17} + 0.232\imath$	$-4.06 \cdot 10^{-16} - 3.4\imath$	$2.65 \cdot 10^{-10} - 3.23\imath$	$-7.0 \cdot 10^{-12} - 3.23\imath$
6.0	$1.21 \cdot 10^{-17} - 0.232\imath$	$3.94 \cdot 10^{-15} + 3.4\imath$	$2.72 \cdot 10^{-10} + 3.23\imath$	$-1.24 \cdot 10^{-12} + 3.23\imath$

Table 21: Case `1a` (`cpv macos`), $r = 6$, Z4: poles.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	$5.55 \cdot 10^{-17} + 3.23\imath$	$-2.65 \cdot 10^{-16} + 0.243\imath$	$1.25 \cdot 10^{-11} - 6.27 \cdot 10^{-11}\imath$	$2.52 \cdot 10^{-13} - 2.27 \cdot 10^{-11}\imath$
2.0	$5.55 \cdot 10^{-17} - 3.23\imath$	$2.35 \cdot 10^{-16} - 0.243\imath$	$1.66 \cdot 10^{-11} + 0.5\imath$	$3.42 \cdot 10^{-13} + 0.5\imath$
3.0	$-1.39 \cdot 10^{-16} + 0.866\imath$	$7.55 \cdot 10^{-16} + 0.908\imath$	$1.68 \cdot 10^{-11} - 0.5\imath$	$3.23 \cdot 10^{-13} - 0.5\imath$
4.0	$-1.39 \cdot 10^{-16} - 0.866\imath$	$-1.43 \cdot 10^{-16} - 0.908\imath$	$4.96 \cdot 10^{-11} + 1.5\imath$	$9.76 \cdot 10^{-13} + 1.5\imath$
5.0	$1.21 \cdot 10^{-17} + 0.232\imath$	$3.11 \cdot 10^{-17} + 3.4\imath$	$5.05 \cdot 10^{-11} - 1.5\imath$	$9.48 \cdot 10^{-13} - 1.5\imath$
6.0	$1.21 \cdot 10^{-17} - 0.232\imath$	$-2.45 \cdot 10^{-15} - 3.4\imath$	$2.31 \cdot 10^{9} + 1.15 \cdot 10^{10}\imath$	$3.47 \cdot 10^{8} + 3.3 \cdot 10^{10}\imath$

Table 22: Case `1a` (`cpv winos`), $r = 6$, Z4: poles.

5.3 Zeros

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.8 \cdot 10^{-11} + 2.97 \cdot 10^{-11}\imath$	$-2.66 \cdot 10^{-13} + 1.23 \cdot 10^{-11}\imath$
2.0	0	$1.1 \cdot 10^{-15} + 7.9 \cdot 10^{-16}\imath$	$2.34 \cdot 10^{-11} - 0.5\imath$	$-7.85 \cdot 10^{-13} - 0.5\imath$
3.0	$1.5\imath$	$1.13 \cdot 10^{-15} - 0.523\imath$	$2.46 \cdot 10^{-11} + 0.5\imath$	$1.08 \cdot 10^{-13} + 0.5\imath$
4.0	$-1.5\imath$	$6.11 \cdot 10^{-16} + 0.523\imath$	$7.02 \cdot 10^{-11} - 1.5\imath$	$-2.43 \cdot 10^{-12} - 1.5\imath$
5.0	$0.5\imath$	$8.46 \cdot 10^{-16} - 1.57\imath$	$7.38 \cdot 10^{-11} + 1.5\imath$	$2.62 \cdot 10^{-13} + 1.5\imath$
6.0	$-0.5\imath$	$5.09 \cdot 10^{-16} + 1.57\imath$	$1.18 \cdot 10^{10} - 1.88 \cdot 10^{10}\imath$	$-1.58 \cdot 10^{9} - 6.51 \cdot 10^{10}\imath$

Table 23: Case `1a` (`cpv macos`), $r = 6$, Z4: zeros.

	Opt., $\sigma_6 = 1.4 \cdot 10^{-07}$	LF, $\sigma_6 = 3.2 \cdot 10^{-07}$	AAA-sign-L200, $\sigma_6 = 1.4 \cdot 10^{-07}$	AAA-sign-L1000, $\sigma_6 = 1.4 \cdot 10^{-07}$
1.0	NaN	NaN	$1.34 \cdot 10^{-11} + 0.232\imath$	$2.75 \cdot 10^{-13} + 0.232\imath$
2.0	0	$1.18 \cdot 10^{-16} - 5.87 \cdot 10^{-17}\imath$	$1.35 \cdot 10^{-11} - 0.232\imath$	$2.66 \cdot 10^{-13} - 0.232\imath$
3.0	$1.5\imath$	$3.18 \cdot 10^{-17} + 0.523\imath$	$2.48 \cdot 10^{-11} + 0.866\imath$	$5.0 \cdot 10^{-13} + 0.866\imath$
4.0	$-1.5\imath$	$-1.45 \cdot 10^{-17} - 0.523\imath$	$2.53 \cdot 10^{-11} - 0.866\imath$	$4.74 \cdot 10^{-13} - 0.866\imath$
5.0	$0.5\imath$	$4.28 \cdot 10^{-16} + 1.57\imath$	$1.86 \cdot 10^{-10} + 3.23\imath$	$3.6 \cdot 10^{-12} + 3.23\imath$
6.0	$-0.5\imath$	$-5.15 \cdot 10^{-16} - 1.57\imath$	$1.88 \cdot 10^{-10} - 3.23\imath$	$3.54 \cdot 10^{-12} - 3.23\imath$

Table 24: Case `1a` (`cpv winos`), $r = 6$, Z4: zeros.

5.4 Poles and zeros visualisation

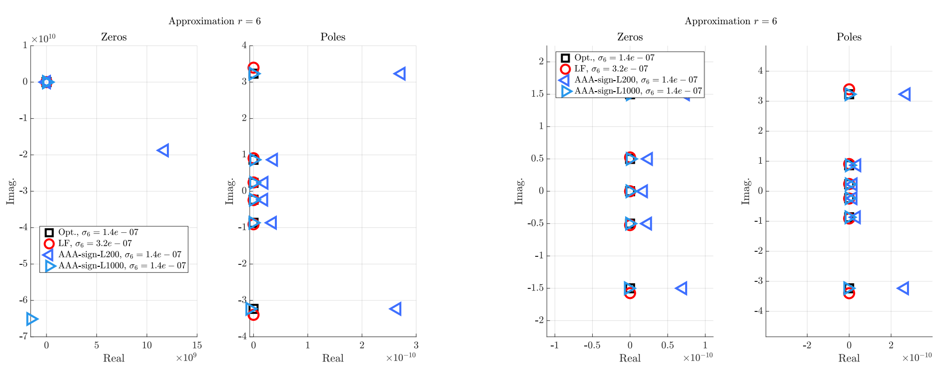

Figure 7: Case `'1a'`, $r = 6$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

5.5 Approximation visualization

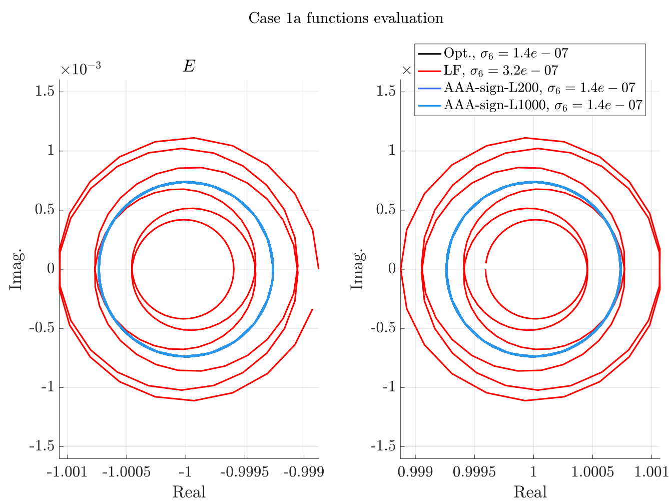

Figure 8: Case '1a', $r = 6$. E and F evaluations.

6 Approximation objective $r = 7$

6.1 Numerator and denominator coefficients

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.5 \cdot 10^{6}$	$-1.8 \cdot 10^{8}$	$-5.6 \cdot 10^{5}$	$-2.6 \cdot 10^{8}$
z^6	6.1	0	7.2	$-1.9 \cdot 10^{-14}$	6.0	$1.1 \cdot 10^{-3}$	6.0	$7.4 \cdot 10^{-4}$
z^5	0	0	$-1.5 \cdot 10^{-13}$	$-5.8 \cdot 10^{-15}$	$-4.0 \cdot 10^{7}$	$-2.8 \cdot 10^{9}$	$-8.8 \cdot 10^{6}$	$-4.0 \cdot 10^{9}$
z^4	23.0	0	36.0	$-7.5 \cdot 10^{-14}$	22.0	0.012	22.0	$8.5 \cdot 10^{-3}$
z^3	0	0	$-3.2 \cdot 10^{-13}$	$-4.4 \cdot 10^{-14}$	$-5.0 \cdot 10^{7}$	$-3.5 \cdot 10^{9}$	$-1.1 \cdot 10^{7}$	$-5.0 \cdot 10^{9}$
z^2	10.0	0	20.0	$-1.2 \cdot 10^{-13}$	10.0	$8.7 \cdot 10^{-3}$	10.0	$6.6 \cdot 10^{-3}$
z	0	0	$-5.3 \cdot 10^{-14}$	$-3.8 \cdot 10^{-14}$	$-7.5 \cdot 10^{6}$	$-5.2 \cdot 10^{8}$	$-1.6 \cdot 10^{6}$	$-7.6 \cdot 10^{8}$
1.0	0.37	0	0.85	$-1.2 \cdot 10^{-14}$	0.35	$4.3 \cdot 10^{-4}$	0.35	$3.4 \cdot 10^{-4}$
z^7	1.0	0	1.0	0	1.0	$1.7 \cdot 10^{-17}$	1.0	0
z^6	0	0	$-2.9 \cdot 10^{-14}$	$-1.6 \cdot 10^{-14}$	$-1.5 \cdot 10^{7}$	$-1.1 \cdot 10^{9}$	$-3.4 \cdot 10^{6}$	$-1.6 \cdot 10^{9}$
z^5	16.0	0	22.0	$-5.2 \cdot 10^{-14}$	16.0	$5.5 \cdot 10^{-3}$	16.0	$3.9 \cdot 10^{-3}$
z^4	0	0	$-3.1 \cdot 10^{-13}$	$1.8 \cdot 10^{-16}$	$-5.8 \cdot 10^{7}$	$-4.0 \cdot 10^{9}$	$-1.3 \cdot 10^{7}$	$-5.8 \cdot 10^{9}$
z^3	20.0	0	35.0	$-1.1 \cdot 10^{-13}$	19.0	0.014	19.0	0.01
z^2	0	0	$-1.8 \cdot 10^{-13}$	$-7.8 \cdot 10^{-14}$	$-2.6 \cdot 10^{7}$	$-1.8 \cdot 10^{9}$	$-5.7 \cdot 10^{6}$	$-2.6 \cdot 10^{9}$
z	3.0	0	6.3	$-6.4 \cdot 10^{-14}$	2.9	$3.0 \cdot 10^{-3}$	2.9	$2.3 \cdot 10^{-3}$
1.0	0	0	$-6.8 \cdot 10^{-15}$	$-3.7 \cdot 10^{-16}$	$-9.3 \cdot 10^{5}$	$-6.5 \cdot 10^{7}$	$-2.0 \cdot 10^{5}$	$-9.4 \cdot 10^{7}$

Table 25: Case `1a` (`cpv macos`), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$		LF, $\sigma_7 = 5.2 \cdot 10^{-08}$		AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$		AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^7	0	0	0	0	$-2.0 \cdot 10^{6}$	$-1.5 \cdot 10^{8}$	$6.3 \cdot 10^{4}$	$-1.7 \cdot 10^{8}$
z^6	6.1	0	7.2	$-1.5 \cdot 10^{-14}$	6.1	$-3.2 \cdot 10^{-5}$	6.1	$4.4 \cdot 10^{-4}$
z^5	0	0	$-4.4 \cdot 10^{-14}$	$1.0 \cdot 10^{-13}$	$-3.1 \cdot 10^{7}$	$-2.3 \cdot 10^{9}$	$9.9 \cdot 10^{5}$	$-2.7 \cdot 10^{9}$
z^4	23.0	0	36.0	$-1.1 \cdot 10^{-13}$	23.0	$-3.5 \cdot 10^{-4}$	23.0	$4.4 \cdot 10^{-3}$
z^3	0	0	$-4.8 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-3.9 \cdot 10^{7}$	$-2.9 \cdot 10^{9}$	$1.2 \cdot 10^{6}$	$-3.4 \cdot 10^{9}$
z^2	10.0	0	20.0	$-6.7 \cdot 10^{-14}$	10.0	$-3.1 \cdot 10^{-4}$	10.0	$3.0 \cdot 10^{-3}$
z	0	0	$-2.2 \cdot 10^{-15}$	$4.1 \cdot 10^{-14}$	$-5.8 \cdot 10^{6}$	$-4.4 \cdot 10^{8}$	$1.9 \cdot 10^{5}$	$-5.1 \cdot 10^{8}$
1.0	0.37	0	0.85	$-9.7 \cdot 10^{-15}$	0.37	$-1.6 \cdot 10^{-5}$	0.37	$1.3 \cdot 10^{-4}$
z^7	1.0	0	1.0	$-5.2 \cdot 10^{-17}$	1.0	0	1.0	0
z^6	0	0	$-7.7 \cdot 10^{-15}$	$1.5 \cdot 10^{-14}$	$-1.2 \cdot 10^{7}$	$-9.0 \cdot 10^{8}$	$3.8 \cdot 10^{5}$	$-1.0 \cdot 10^{9}$
z^5	16.0	0	22.0	$-6.0 \cdot 10^{-14}$	16.0	$-1.7 \cdot 10^{-4}$	16.0	$2.1 \cdot 10^{-3}$
z^4	0	0	$-7.1 \cdot 10^{-14}$	$2.0 \cdot 10^{-13}$	$-4.5 \cdot 10^{7}$	$-3.4 \cdot 10^{9}$	$1.4 \cdot 10^{6}$	$-3.9 \cdot 10^{9}$
z^3	20.0	0	35.0	$-1.1 \cdot 10^{-13}$	20.0	$-4.2 \cdot 10^{-4}$	20.0	$4.9 \cdot 10^{-3}$
z^2	0	0	$-1.6 \cdot 10^{-14}$	$1.2 \cdot 10^{-13}$	$-2.0 \cdot 10^{7}$	$-1.5 \cdot 10^{9}$	$6.4 \cdot 10^{5}$	$-1.7 \cdot 10^{9}$
z	3.0	0	6.3	$-3.5 \cdot 10^{-14}$	3.0	$-1.1 \cdot 10^{-4}$	3.0	$9.7 \cdot 10^{-4}$
1.0	0	0	$1.0 \cdot 10^{-15}$	$7.7 \cdot 10^{-15}$	$-7.2 \cdot 10^{5}$	$-5.4 \cdot 10^{7}$	$2.3 \cdot 10^{4}$	$-6.2 \cdot 10^{7}$

Table 26: Case `1a` (`cpv winos`), $r = 7$, Z4: numerator (first lines) and denominator (last lines) coefficients.

6.2 Poles

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$1.08 \cdot 10^{-15} + 5.86 \cdot 10^{-17}\imath$	$1.63 \cdot 10^{-11} + 0.198\imath$	$5.75 \cdot 10^{-12} + 0.198\imath$
2.0	$-7.63 \cdot 10^{-17} + 3.79\imath$	$-1.46 \cdot 10^{-16} - 0.458\imath$	$5.67 \cdot 10^{-12} - 0.198\imath$	$-2.73 \cdot 10^{-12} - 0.198\imath$
3.0	$-7.63 \cdot 10^{-17} - 3.79\imath$	$3.4 \cdot 10^{-15} + 0.458\imath$	$3.53 \cdot 10^{-11} + 0.691\imath$	$1.68 \cdot 10^{-11} + 0.691\imath$
4.0	$2.78 \cdot 10^{-17} + 1.09\imath$	$3.32 \cdot 10^{-15} + 1.23\imath$	$-1.59 \cdot 10^{-12} - 0.691\imath$	$-1.26 \cdot 10^{-11} - 0.691\imath$
5.0	$2.78 \cdot 10^{-17} - 1.09\imath$	$5.36 \cdot 10^{-15} - 1.23\imath$	$1.01 \cdot 10^{-10} + 1.8\imath$	$4.36 \cdot 10^{-11} + 1.8\imath$
6.0	$7.63 \cdot 10^{-17} + 0.417\imath$	$1.82 \cdot 10^{-15} - 4.47\imath$	$6.14 \cdot 10^{-12} - 1.8\imath$	$-3.18 \cdot 10^{-11} - 1.8\imath$
7.0	$7.63 \cdot 10^{-17} - 0.417\imath$	$1.53 \cdot 10^{-14} + 4.47\imath$	$1.54 \cdot 10^{7} + 1.07 \cdot 10^{9}\imath$	$3.38 \cdot 10^{6} + 1.55 \cdot 10^{9}\imath$

Table 27: Case `1a` (`cpv macos`), $r = 7$, Z4: poles.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	0	$-1.61 \cdot 10^{-16} - 1.23 \cdot 10^{-15}\imath$	$1.16 \cdot 10^{-11} + 0.198\imath$	$-2.97 \cdot 10^{-12} + 0.198\imath$
2.0	$-7.63 \cdot 10^{-17} + 3.79\imath$	$6.42 \cdot 10^{-16} + 0.458\imath$	$1.16 \cdot 10^{-11} - 0.198\imath$	$2.94 \cdot 10^{-12} - 0.198\imath$
3.0	$-7.63 \cdot 10^{-17} - 3.79\imath$	$-4.86 \cdot 10^{-16} - 0.458\imath$	$1.81 \cdot 10^{-11} + 0.691\imath$	$-1.0 \cdot 10^{-11} + 0.691\imath$
4.0	$2.78 \cdot 10^{-17} + 1.09\imath$	$1.24 \cdot 10^{-15} + 1.23\imath$	$1.81 \cdot 10^{-11} - 0.691\imath$	$9.84 \cdot 10^{-12} - 0.691\imath$
5.0	$2.78 \cdot 10^{-17} - 1.09\imath$	$1.68 \cdot 10^{-15} - 1.23\imath$	$5.9 \cdot 10^{-11} + 1.8\imath$	$-2.56 \cdot 10^{-11} + 1.8\imath$
6.0	$7.63 \cdot 10^{-17} + 0.417\imath$	$8.32 \cdot 10^{-15} + 4.47\imath$	$5.86 \cdot 10^{-11} - 1.8\imath$	$2.36 \cdot 10^{-11} - 1.8\imath$
7.0	$7.63 \cdot 10^{-17} - 0.417\imath$	$-9.17 \cdot 10^{-16} - 4.47\imath$	$1.2 \cdot 10^{7} + 9.01 \cdot 10^{8}\imath$	$-3.8 \cdot 10^{5} + 1.04 \cdot 10^{9}\imath$

Table 28: Case `1a` (`cpv winos`), $r = 7$, Z4: poles.

6.3 Zeros

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.05 \cdot 10^{-11} - 6.74 \cdot 10^{-10}\imath$	$1.46 \cdot 10^{-12} - 4.66 \cdot 10^{-10}\imath$
2.0	$1.73 \cdot 10^{-17} + 1.8\imath$	$2.11 \cdot 10^{-16} - 0.216\imath$	$2.4 \cdot 10^{-11} + 0.417\imath$	$1.06 \cdot 10^{-11} + 0.417\imath$
3.0	$1.73 \cdot 10^{-17} - 1.8\imath$	$2.13 \cdot 10^{-15} + 0.216\imath$	$1.61 \cdot 10^{-12} - 0.417\imath$	$-7.23 \cdot 10^{-12} - 0.417\imath$
4.0	$5.55 \cdot 10^{-17} + 0.691\imath$	$1.4 \cdot 10^{-15} - 0.768\imath$	$5.5 \cdot 10^{-11} + 1.09\imath$	$2.6 \cdot 10^{-11} + 1.09\imath$
5.0	$5.55 \cdot 10^{-17} - 0.691\imath$	$4.64 \cdot 10^{-15} + 0.768\imath$	$-2.6 \cdot 10^{-12} - 1.09\imath$	$-1.99 \cdot 10^{-11} - 1.09\imath$
6.0	$1.21 \cdot 10^{-17} + 0.198\imath$	$5.1 \cdot 10^{-15} + 2.07\imath$	$3.02 \cdot 10^{-10} + 3.79\imath$	$1.01 \cdot 10^{-10} + 3.79\imath$
7.0	$1.21 \cdot 10^{-17} - 0.198\imath$	$7.22 \cdot 10^{-15} - 2.07\imath$	$1.02 \cdot 10^{-10} - 3.79\imath$	$-5.76 \cdot 10^{-11} - 3.79\imath$

Table 29: Case `1a` (`cpv macos`), $r = 7$, Z4: zeros.

	Opt., $\sigma_7 = 9.8 \cdot 10^{-09}$	LF, $\sigma_7 = 5.2 \cdot 10^{-08}$	AAA-sign-L200, $\sigma_7 = 9.8 \cdot 10^{-09}$	AAA-sign-L1000, $\sigma_7 = 9.8 \cdot 10^{-09}$
1.0	NaN	NaN	$1.11 \cdot 10^{-11} - 8.34 \cdot 10^{-10}\imath$	$-2.09 \cdot 10^{-14} - 7.35 \cdot 10^{-10}\imath$
2.0	$1.73 \cdot 10^{-17} + 1.8\imath$	$8.87 \cdot 10^{-16} + 0.216\imath$	$1.36 \cdot 10^{-11} + 0.417\imath$	$-6.16 \cdot 10^{-12} + 0.417\imath$
3.0	$1.73 \cdot 10^{-17} - 1.8\imath$	$-8.68 \cdot 10^{-16} - 0.216\imath$	$1.36 \cdot 10^{-11} - 0.417\imath$	$6.12 \cdot 10^{-12} - 0.417\imath$
4.0	$5.55 \cdot 10^{-17} + 0.691\imath$	$2.39 \cdot 10^{-16} + 0.768\imath$	$2.86 \cdot 10^{-11} + 1.09\imath$	$-1.55 \cdot 10^{-11} + 1.09\imath$
5.0	$5.55 \cdot 10^{-17} - 0.691\imath$	$4.76 \cdot 10^{-16} - 0.768\imath$	$2.84 \cdot 10^{-11} - 1.09\imath$	$1.49 \cdot 10^{-11} - 1.09\imath$
6.0	$1.21 \cdot 10^{-17} + 0.198\imath$	$3.76 \cdot 10^{-15} + 2.07\imath$	$2.24 \cdot 10^{-10} + 3.79\imath$	$-5.59 \cdot 10^{-11} + 3.79\imath$
7.0	$1.21 \cdot 10^{-17} - 0.198\imath$	$1.55 \cdot 10^{-15} - 2.07\imath$	$2.23 \cdot 10^{-10} - 3.79\imath$	$4.61 \cdot 10^{-11} - 3.79\imath$

Table 30: Case `1a` (`cpv winos`), $r = 7$, Z4: zeros.

6.4 Poles and zeros visualisation

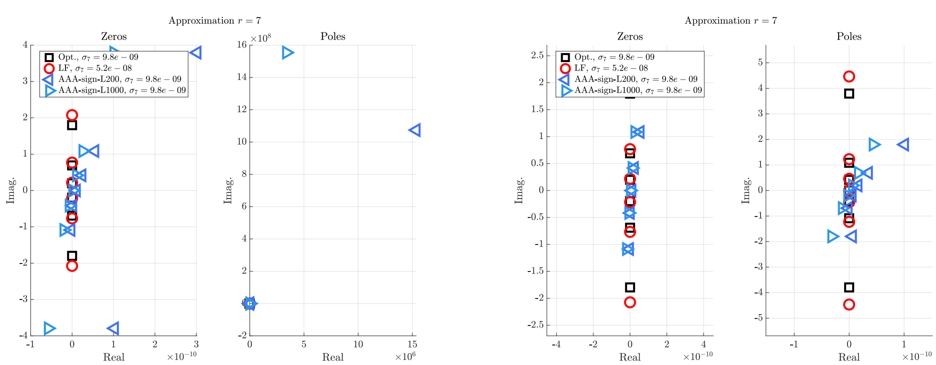

Figure 9: Case `'1a'`, $r = 7$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

6.5 Approximation visualization

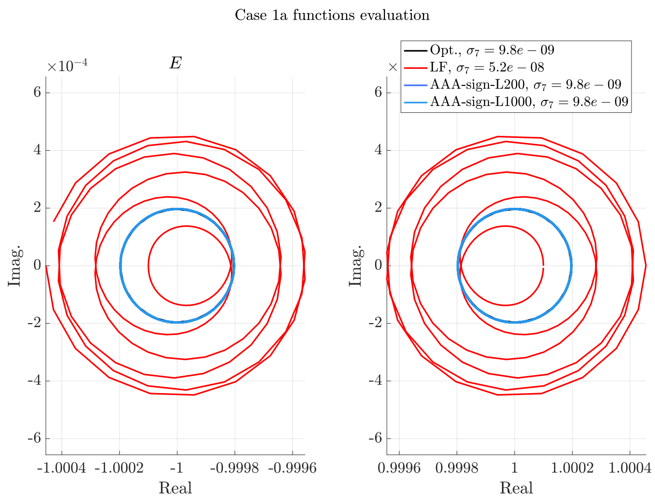

Figure 10: Case `'1a'`, $r = 7$. E and F evaluations.

7 Approximation objective $r = 8$

7.1 Numerator and denominator coefficients

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-8.7 \cdot 10^5$	$1.7 \cdot 10^8$	$-5.1 \cdot 10^6$	$-7.9 \cdot 10^8$
z^7	6.9	0	7.2	$-1.9 \cdot 10^{-14}$	7.0	$-4.9 \cdot 10^{-4}$	7.0	$-3.4 \cdot 10^{-3}$
z^6	0	0	$-5.3 \cdot 10^{-14}$	$-1.4 \cdot 10^{-13}$	$-1.8 \cdot 10^7$	$3.7 \cdot 10^9$	$-1.1 \cdot 10^8$	$-1.7 \cdot 10^{10}$
z^5	36.0	0	41.0	$-7.4 \cdot 10^{-14}$	37.0	$-9.8 \cdot 10^{-3}$	37.0	-0.046
z^4	0	0	$7.7 \cdot 10^{-14}$	$-3.2 \cdot 10^{-13}$	$-3.4 \cdot 10^7$	$6.9 \cdot 10^9$	$-2.0 \cdot 10^8$	$-3.1 \cdot 10^{10}$
z^3	27.0	0	33.0	$-2.2 \cdot 10^{-14}$	28.0	-0.014	28.0	-0.051
z^2	0	0	$1.6 \cdot 10^{-13}$	$-7.9 \cdot 10^{-14}$	$-1.0 \cdot 10^7$	$2.1 \cdot 10^9$	$-6.0 \cdot 10^7$	$-9.3 \cdot 10^9$
z	2.9	0	3.7	$3.3 \cdot 10^{-14}$	3.0	$-2.4 \cdot 10^{-3}$	3.1	$-7.0 \cdot 10^{-3}$
1.0	0	0	$1.0 \cdot 10^{-14}$	$1.0 \cdot 10^{-14}$	$-2.8 \cdot 10^5$	$5.5 \cdot 10^7$	$-1.6 \cdot 10^6$	$-2.5 \cdot 10^8$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-1.4 \cdot 10^{-14}$	$-4.1 \cdot 10^{-14}$	$-6.0 \cdot 10^6$	$1.2 \cdot 10^9$	$-3.5 \cdot 10^7$	$-5.4 \cdot 10^9$
z^6	21.0	0	23.0	$-5.5 \cdot 10^{-14}$	21.0	$-3.4 \cdot 10^{-3}$	21.0	-0.019
z^5	0	0	$-5.6 \cdot 10^{-14}$	$-2.7 \cdot 10^{-13}$	$-3.2 \cdot 10^7$	$6.3 \cdot 10^9$	$-1.9 \cdot 10^8$	$-2.9 \cdot 10^{10}$
z^4	39.0	0	46.0	$-6.5 \cdot 10^{-14}$	40.0	-0.015	40.0	-0.062
z^3	0	0	$1.9 \cdot 10^{-13}$	$-2.4 \cdot 10^{-13}$	$-2.4 \cdot 10^7$	$4.7 \cdot 10^9$	$-1.4 \cdot 10^8$	$-2.1 \cdot 10^{10}$
z^2	12.0	0	15.0	$3.1 \cdot 10^{-14}$	12.0	$-8.0 \cdot 10^{-3}$	12.0	-0.025
z	0	0	$6.2 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$-2.5 \cdot 10^6$	$5.1 \cdot 10^8$	$-1.5 \cdot 10^7$	$-2.3 \cdot 10^9$
1.0	0.32	0	0.42	$8.3 \cdot 10^{-15}$	0.33	$-3.2 \cdot 10^{-4}$	0.33	$-8.5 \cdot 10^{-4}$

Table 31: Case `1a` (`cpv macos`), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$		LF, $\sigma_8 = 1.6 \cdot 10^{-09}$		AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$		AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^8	0	0	0	0	$-5.8 \cdot 10^{-11}$	$-1.2 \cdot 10^{-8}$	$2.5 \cdot 10^{-12}$	$1.4 \cdot 10^{-9}$
z^7	6.9	0	7.2	$-1.5 \cdot 10^{-14}$	6.9	$-2.1 \cdot 10^{-10}$	6.9	$-5.3 \cdot 10^{-11}$
z^6	0	0	$-5.0 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.2 \cdot 10^{-9}$	$-2.6 \cdot 10^{-7}$	$6.3 \cdot 10^{-11}$	$2.9 \cdot 10^{-8}$
z^5	36.0	0	41.0	$-2.6 \cdot 10^{-14}$	36.0	$-3.3 \cdot 10^{-9}$	36.0	$-8.3 \cdot 10^{-10}$
z^4	0	0	$-1.7 \cdot 10^{-13}$	$1.5 \cdot 10^{-13}$	$-2.1 \cdot 10^{-9}$	$-4.8 \cdot 10^{-7}$	$1.3 \cdot 10^{-10}$	$5.5 \cdot 10^{-8}$
z^3	27.0	0	33.0	$1.7 \cdot 10^{-13}$	27.0	$-4.1 \cdot 10^{-9}$	27.0	$-1.0 \cdot 10^{-9}$
z^2	0	0	$-8.5 \cdot 10^{-14}$	$9.5 \cdot 10^{-14}$	$-6.3 \cdot 10^{-10}$	$-1.5 \cdot 10^{-7}$	$4.3 \cdot 10^{-11}$	$1.6 \cdot 10^{-8}$
z	2.9	0	3.7	$4.3 \cdot 10^{-14}$	2.9	$-6.2 \cdot 10^{-10}$	2.9	$-1.6 \cdot 10^{-10}$
1.0	0	0	$-2.2 \cdot 10^{-15}$	$4.0 \cdot 10^{-15}$	$-1.7 \cdot 10^{-11}$	$-3.9 \cdot 10^{-9}$	$1.3 \cdot 10^{-12}$	$4.3 \cdot 10^{-10}$
z^8	1.0	0	1.0	0	1.0	0	1.0	0
z^7	0	0	$-7.1 \cdot 10^{-15}$	$-2.0 \cdot 10^{-15}$	$-3.9 \cdot 10^{-10}$	$-8.4 \cdot 10^{-8}$	$1.9 \cdot 10^{-11}$	$9.7 \cdot 10^{-9}$
z^6	21.0	0	23.0	$-4.0 \cdot 10^{-14}$	21.0	$-1.3 \cdot 10^{-9}$	21.0	$-3.2 \cdot 10^{-10}$
z^5	0	0	$-1.2 \cdot 10^{-13}$	$9.2 \cdot 10^{-14}$	$-2.0 \cdot 10^{-9}$	$-4.5 \cdot 10^{-7}$	$1.2 \cdot 10^{-10}$	$5.1 \cdot 10^{-8}$
z^4	39.0	0	46.0	$9.0 \cdot 10^{-14}$	39.0	$-4.8 \cdot 10^{-9}$	39.0	$-1.2 \cdot 10^{-9}$
z^3	0	0	$-1.6 \cdot 10^{-13}$	$1.6 \cdot 10^{-13}$	$-1.5 \cdot 10^{-9}$	$-3.4 \cdot 10^{-7}$	$9.7 \cdot 10^{-11}$	$3.8 \cdot 10^{-8}$
z^2	12.0	0	15.0	$1.2 \cdot 10^{-13}$	12.0	$-2.1 \cdot 10^{-9}$	12.0	$-5.4 \cdot 10^{-10}$
z	0	0	$-2.1 \cdot 10^{-14}$	$3.3 \cdot 10^{-14}$	$-1.5 \cdot 10^{-10}$	$-3.6 \cdot 10^{-8}$	$1.1 \cdot 10^{-11}$	$4.0 \cdot 10^{-9}$
1.0	0.32	0	0.42	$6.7 \cdot 10^{-15}$	0.32	$-7.6 \cdot 10^{-11}$	0.32	$-1.9 \cdot 10^{-11}$

Table 32: Case `1a` (`cpv winos`), $r = 8$, Z4: numerator (first lines) and denominator (last lines) coefficients.

7.2 Poles

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$6.66 \cdot 10^{-16} + 4.35\imath$	$-4.06 \cdot 10^{-15} + 0.178\imath$	$3.84 \cdot 10^{-12} + 6.42 \cdot 10^{-10}\imath$	$5.64 \cdot 10^{-13} - 1.45 \cdot 10^{-10}\imath$
2.0	$6.66 \cdot 10^{-16} - 4.35\imath$	$-6.53 \cdot 10^{-16} - 0.178\imath$	$-4.23 \cdot 10^{-12} - 0.359\imath$	$-2.48 \cdot 10^{-12} + 0.359\imath$
3.0	$2.22 \cdot 10^{-16} + 1.3\imath$	$-7.33 \cdot 10^{-17} - 0.599\imath$	$1.28 \cdot 10^{-11} + 0.359\imath$	$3.91 \cdot 10^{-12} - 0.359\imath$
4.0	$2.22 \cdot 10^{-16} - 1.3\imath$	$-1.58 \cdot 10^{-15} + 0.599\imath$	$-1.39 \cdot 10^{-11} - 0.866\imath$	$-6.57 \cdot 10^{-12} + 0.866\imath$
5.0	$1.39 \cdot 10^{-17} + 0.579\imath$	$4.61 \cdot 10^{-15} - 1.34\imath$	$2.73 \cdot 10^{-11} + 0.866\imath$	$9.26 \cdot 10^{-12} - 0.866\imath$
6.0	$1.39 \cdot 10^{-17} - 0.579\imath$	$3.38 \cdot 10^{-15} + 1.34\imath$	$-2.86 \cdot 10^{-11} - 2.09\imath$	$-1.37 \cdot 10^{-11} + 2.09\imath$
7.0	$1.47 \cdot 10^{-17} + 0.172\imath$	$-4.63 \cdot 10^{-16} - 4.52\imath$	$7.16 \cdot 10^{-11} + 2.09\imath$	$2.47 \cdot 10^{-11} - 2.09\imath$
8.0	$1.47 \cdot 10^{-17} - 0.172\imath$	$1.25 \cdot 10^{-14} + 4.52\imath$	$6.04 \cdot 10^6 - 1.21 \cdot 10^9\imath$	$3.54 \cdot 10^7 + 5.44 \cdot 10^9\imath$

Table 33: Case `1a` (`cpv macos`), $r = 8$, Z4: poles.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	$7.22 \cdot 10^{-16} + 4.35\imath$	$1.37 \cdot 10^{-17} + 0.178\imath$	$7.54 \cdot 10^{-13} - 0.172\imath$	$8.64 \cdot 10^{-13} + 0.172\imath$
2.0	$7.22 \cdot 10^{-16} - 4.35\imath$	$1.31 \cdot 10^{-15} - 0.178\imath$	$1.1 \cdot 10^{-11} + 0.172\imath$	$-1.73 \cdot 10^{-12} - 0.172\imath$
3.0	$1.94 \cdot 10^{-16} + 1.3\imath$	$-1.07 \cdot 10^{-15} + 0.599\imath$	$-9.06 \cdot 10^{-12} - 0.579\imath$	$3.82 \cdot 10^{-12} + 0.579\imath$
4.0	$1.94 \cdot 10^{-16} - 1.3\imath$	$3.39 \cdot 10^{-15} - 0.599\imath$	$2.57 \cdot 10^{-11} + 0.579\imath$	$-4.98 \cdot 10^{-12} - 0.579\imath$
5.0	$-4.16 \cdot 10^{-17} + 0.579\imath$	$-1.82 \cdot 10^{-15} + 1.34\imath$	$-2.0 \cdot 10^{-11} - 1.3\imath$	$8.73 \cdot 10^{-12} + 1.3\imath$
6.0	$-4.16 \cdot 10^{-17} - 0.579\imath$	$3.47 \cdot 10^{-15} - 1.34\imath$	$5.85 \cdot 10^{-11} + 1.3\imath$	$-1.1 \cdot 10^{-11} - 1.3\imath$
7.0	$1.73 \cdot 10^{-18} + 0.172\imath$	$-5.1 \cdot 10^{-15} - 4.52\imath$	$2.98 \cdot 10^{-11} - 4.35\imath$	$2.57 \cdot 10^{-11} + 4.35\imath$
8.0	$1.73 \cdot 10^{-18} - 0.172\imath$	$6.85 \cdot 10^{-15} + 4.52\imath$	$2.95 \cdot 10^{-10} + 4.35\imath$	$-4.05 \cdot 10^{-11} - 4.35\imath$

Table 34: Case `1a` (`cpv winos`), $r = 8$, Z4: poles.

7.3 Zeros

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$-1.46 \cdot 10^{-13} - 0.172\imath$	$-9.08 \cdot 10^{-13} + 0.172\imath$
2.0	0	$-2.76 \cdot 10^{-15} - 2.73 \cdot 10^{-15}\imath$	$8.04 \cdot 10^{-12} + 0.172\imath$	$2.11 \cdot 10^{-12} - 0.172\imath$
3.0	$1.39 \cdot 10^{-16} + 2.09\imath$	$4.64 \cdot 10^{-16} - 0.371\imath$	$-8.67 \cdot 10^{-12} - 0.579\imath$	$-4.32 \cdot 10^{-12} + 0.579\imath$
4.0	$1.39 \cdot 10^{-16} - 2.09\imath$	$-3.67 \cdot 10^{-15} + 0.371\imath$	$1.88 \cdot 10^{-11} + 0.579\imath$	$6.15 \cdot 10^{-12} - 0.579\imath$
5.0	$-5.55 \cdot 10^{-17} + 0.866\imath$	$8.79 \cdot 10^{-16} - 0.897\imath$	$-2.05 \cdot 10^{-11} - 1.3\imath$	$-9.53 \cdot 10^{-12} + 1.3\imath$
6.0	$-5.55 \cdot 10^{-17} - 0.866\imath$	$1.64 \cdot 10^{-15} + 0.897\imath$	$4.14 \cdot 10^{-11} + 1.3\imath$	$1.43 \cdot 10^{-11} - 1.3\imath$
7.0	$-6.94 \cdot 10^{-18} + 0.359\imath$	$6.16 \cdot 10^{-15} - 2.17\imath$	$-2.26 \cdot 10^{-11} - 4.35\imath$	$-1.71 \cdot 10^{-11} + 4.35\imath$
8.0	$-6.94 \cdot 10^{-18} - 0.359\imath$	$4.41 \cdot 10^{-15} + 2.17\imath$	$1.86 \cdot 10^{-10} + 4.35\imath$	$6.27 \cdot 10^{-11} - 4.35\imath$

Table 35: Case `1a` (`cpv macos`), $r = 8$, Z4: zeros.

	Opt., $\sigma_8 = 7.1 \cdot 10^{-10}$	LF, $\sigma_8 = 1.6 \cdot 10^{-09}$	AAA-sign-L200, $\sigma_8 = 7.1 \cdot 10^{-10}$	AAA-sign-L1000, $\sigma_8 = 7.1 \cdot 10^{-10}$
1.0	NaN	NaN	$5.66 \cdot 10^{-12} + 1.34 \cdot 10^{-9}\imath$	$-4.28 \cdot 10^{-13} - 1.47 \cdot 10^{-10}\imath$
2.0	0	$5.49 \cdot 10^{-16} - 1.11 \cdot 10^{-15}\imath$	$-4.07 \cdot 10^{-12} - 0.359\imath$	$2.24 \cdot 10^{-12} + 0.359\imath$
3.0	$1.39 \cdot 10^{-16} + 2.09\imath$	$-4.33 \cdot 10^{-17} + 0.371\imath$	$1.74 \cdot 10^{-11} + 0.359\imath$	$-3.19 \cdot 10^{-12} - 0.359\imath$
4.0	$1.39 \cdot 10^{-16} - 2.09\imath$	$2.25 \cdot 10^{-15} - 0.371\imath$	$-1.44 \cdot 10^{-11} - 0.866\imath$	$5.83 \cdot 10^{-12} + 0.866\imath$
5.0	$-5.55 \cdot 10^{-17} + 0.866\imath$	$-2.49 \cdot 10^{-15} + 0.897\imath$	$3.78 \cdot 10^{-11} + 0.866\imath$	$-7.39 \cdot 10^{-12} - 0.866\imath$
6.0	$-5.55 \cdot 10^{-17} - 0.866\imath$	$3.87 \cdot 10^{-15} - 0.897\imath$	$-2.21 \cdot 10^{-11} - 2.09\imath$	$1.38 \cdot 10^{-11} + 2.09\imath$
7.0	$-6.94 \cdot 10^{-18} + 0.359\imath$	$8.84 \cdot 10^{-16} + 2.17\imath$	$1.05 \cdot 10^{-10} + 2.09\imath$	$-1.81 \cdot 10^{-11} - 2.09\imath$
8.0	$-6.94 \cdot 10^{-18} - 0.359\imath$	$2.13 \cdot 10^{-15} - 2.17\imath$	$2.7 \cdot 10^{6} - 5.7 \cdot 10^{8}\imath$	$-8.64 \cdot 10^{6} + 4.91 \cdot 10^{9}\imath$

Table 36: Case `1a` (`cpv winos`), $r = 8$, Z4: zeros.

7.4 Poles and zeros visualisation

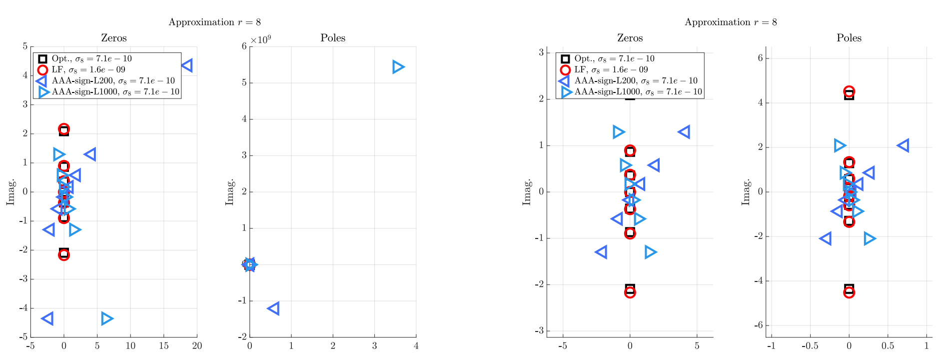

Figure 11: Case '1a', $r = 8$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

7.5 Approximation visualization

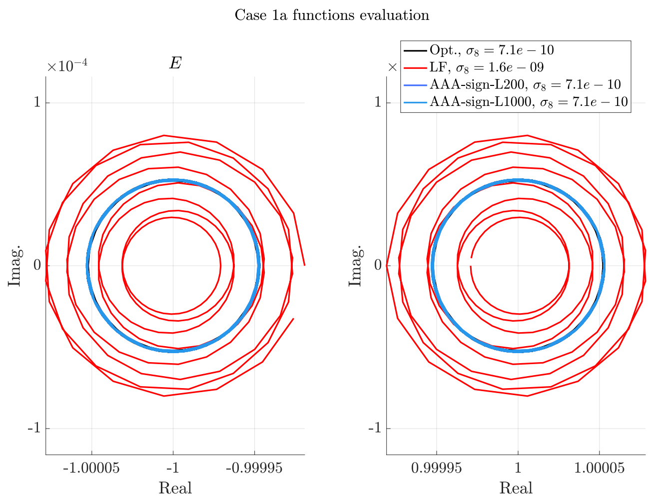

Figure 12: Case `'1a'`, $r = 8$. E and F evaluations.

8 Approximation objective $r = 9$

8.1 Numerator and denominator coefficients

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$-2.1 \cdot 10^{3}$	$1.4 \cdot 10^{6}$	-940.0	$-5.1 \cdot 10^{5}$
z^8	7.8	0	8.9	$5.1 \cdot 10^{-14}$	7.8	$-8.0 \cdot 10^{-4}$	7.8	$3.7 \cdot 10^{-4}$
z^7	0	0	$9.2 \cdot 10^{-13}$	$1.1 \cdot 10^{-12}$	$-5.7 \cdot 10^{4}$	$3.7 \cdot 10^{7}$	$-2.5 \cdot 10^{4}$	$-1.4 \cdot 10^{7}$
z^6	55.0	0	78.0	$1.6 \cdot 10^{-13}$	54.0	-0.016	55.0	$7.1 \cdot 10^{-3}$
z^5	0	0	$4.0 \cdot 10^{-12}$	$6.7 \cdot 10^{-12}$	$-1.5 \cdot 10^{5}$	$9.8 \cdot 10^{7}$	$-6.6 \cdot 10^{4}$	$-3.6 \cdot 10^{7}$
z^4	61.0	0	110.0	$3.2 \cdot 10^{-13}$	60.0	-0.027	62.0	0.012
z^3	0	0	$2.2 \cdot 10^{-12}$	$4.8 \cdot 10^{-12}$	$-7.4 \cdot 10^{4}$	$4.9 \cdot 10^{7}$	$-3.3 \cdot 10^{4}$	$-1.8 \cdot 10^{7}$
z^2	13.0	0	27.0	$3.1 \cdot 10^{-13}$	13.0	$-7.6 \cdot 10^{-3}$	13.0	$3.1 \cdot 10^{-3}$
z	0	0	$6.9 \cdot 10^{-14}$	$5.3 \cdot 10^{-13}$	$-6.0 \cdot 10^{3}$	$3.9 \cdot 10^{6}$	$-2.7 \cdot 10^{3}$	$-1.5 \cdot 10^{6}$
1.0	0.27	0	0.64	$-4.9 \cdot 10^{-17}$	0.27	$-1.9 \cdot 10^{-4}$	0.28	$7.3 \cdot 10^{-5}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$1.5 \cdot 10^{-13}$	$1.3 \cdot 10^{-13}$	$-1.6 \cdot 10^{4}$	$1.1 \cdot 10^{7}$	$-7.3 \cdot 10^{3}$	$-4.0 \cdot 10^{6}$
z^7	27.0	0	35.0	$1.7 \cdot 10^{-13}$	27.0	$-5.3 \cdot 10^{-3}$	27.0	$2.4 \cdot 10^{-3}$
z^6	0	0	$2.5 \cdot 10^{-12}$	$3.7 \cdot 10^{-12}$	$-1.1 \cdot 10^{5}$	$7.5 \cdot 10^{7}$	$-5.1 \cdot 10^{4}$	$-2.8 \cdot 10^{7}$
z^5	71.0	0	110.0	$1.0 \cdot 10^{-13}$	70.0	-0.026	72.0	0.012
z^4	0	0	$3.8 \cdot 10^{-12}$	$7.1 \cdot 10^{-12}$	$-1.3 \cdot 10^{5}$	$8.5 \cdot 10^{7}$	$-5.8 \cdot 10^{4}$	$-3.2 \cdot 10^{7}$
z^3	35.0	0	67.0	$5.1 \cdot 10^{-13}$	35.0	-0.018	36.0	$7.7 \cdot 10^{-3}$
z^2	0	0	$6.6 \cdot 10^{-13}$	$2.1 \cdot 10^{-12}$	$-2.8 \cdot 10^{4}$	$1.8 \cdot 10^{7}$	$-1.2 \cdot 10^{4}$	$-6.8 \cdot 10^{6}$
z	2.8	0	6.2	$6.5 \cdot 10^{-14}$	2.8	$-1.8 \cdot 10^{-3}$	2.9	$7.3 \cdot 10^{-4}$
1.0	0	0	$-6.9 \cdot 10^{-15}$	$6.8 \cdot 10^{-14}$	-570.0	$3.8 \cdot 10^{5}$	-260.0	$-1.4 \cdot 10^{5}$

Table 37: Case `1a` (`cpv macos`), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$		LF, $\sigma_9 = 2.7 \cdot 10^{-10}$		AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$		AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^9	0	0	0	0	$2.5 \cdot 10^{-9}$	$-2.9 \cdot 10^{-6}$	$-3.5 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$
z^8	7.8	0	8.9	$-2.6 \cdot 10^{-13}$	7.8	$3.9 \cdot 10^{-8}$	7.8	$1.9 \cdot 10^{-9}$
z^7	0	0	$3.5 \cdot 10^{-13}$	$-1.3 \cdot 10^{-13}$	$7.7 \cdot 10^{-8}$	$-7.9 \cdot 10^{-5}$	$-8.9 \cdot 10^{-9}$	$-2.1 \cdot 10^{-5}$
z^6	55.0	0	78.0	$-4.4 \cdot 10^{-12}$	55.0	$8.3 \cdot 10^{-7}$	55.0	$4.1 \cdot 10^{-8}$
z^5	0	0	$2.0 \cdot 10^{-12}$	$3.8 \cdot 10^{-13}$	$2.2 \cdot 10^{-7}$	$-2.1 \cdot 10^{-4}$	$-2.2 \cdot 10^{-8}$	$-5.6 \cdot 10^{-5}$
z^4	61.0	0	110.0	$-7.9 \cdot 10^{-12}$	61.0	$1.6 \cdot 10^{-6}$	61.0	$7.7 \cdot 10^{-8}$
z^3	0	0	$1.7 \cdot 10^{-12}$	$6.9 \cdot 10^{-13}$	$1.2 \cdot 10^{-7}$	$-1.0 \cdot 10^{-4}$	$-1.1 \cdot 10^{-8}$	$-2.8 \cdot 10^{-5}$
z^2	13.0	0	27.0	$-2.4 \cdot 10^{-12}$	13.0	$4.7 \cdot 10^{-7}$	13.0	$2.3 \cdot 10^{-8}$
z	0	0	$2.1 \cdot 10^{-13}$	$1.2 \cdot 10^{-13}$	$9.6 \cdot 10^{-9}$	$-8.2 \cdot 10^{-6}$	$-8.4 \cdot 10^{-10}$	$-2.2 \cdot 10^{-6}$
1.0	0.27	0	0.64	$-6.1 \cdot 10^{-14}$	0.27	$1.3 \cdot 10^{-8}$	0.27	$6.3 \cdot 10^{-10}$
z^9	1.0	0	1.0	0	1.0	0	1.0	0
z^8	0	0	$5.0 \cdot 10^{-14}$	$-5.1 \cdot 10^{-14}$	$2.1 \cdot 10^{-8}$	$-2.3 \cdot 10^{-5}$	$-2.6 \cdot 10^{-9}$	$-6.1 \cdot 10^{-6}$
z^7	27.0	0	35.0	$-1.6 \cdot 10^{-12}$	27.0	$2.7 \cdot 10^{-7}$	27.0	$1.3 \cdot 10^{-8}$
z^6	0	0	$1.1 \cdot 10^{-12}$	$-4.6 \cdot 10^{-14}$	$1.6 \cdot 10^{-7}$	$-1.6 \cdot 10^{-4}$	$-1.8 \cdot 10^{-8}$	$-4.3 \cdot 10^{-5}$
z^5	71.0	0	110.0	$-7.4 \cdot 10^{-12}$	71.0	$1.4 \cdot 10^{-6}$	71.0	$7.1 \cdot 10^{-8}$
z^4	0	0	$2.3 \cdot 10^{-12}$	$7.6 \cdot 10^{-13}$	$2.0 \cdot 10^{-7}$	$-1.8 \cdot 10^{-4}$	$-1.9 \cdot 10^{-8}$	$-4.8 \cdot 10^{-5}$
z^3	35.0	0	67.0	$-5.6 \cdot 10^{-12}$	35.0	$1.1 \cdot 10^{-6}$	35.0	$5.4 \cdot 10^{-8}$
z^2	0	0	$8.0 \cdot 10^{-13}$	$3.8 \cdot 10^{-13}$	$4.4 \cdot 10^{-8}$	$-3.8 \cdot 10^{-5}$	$-3.9 \cdot 10^{-9}$	$-1.0 \cdot 10^{-5}$
z	2.8	0	6.2	$-5.9 \cdot 10^{-13}$	2.8	$1.2 \cdot 10^{-7}$	2.8	$5.8 \cdot 10^{-9}$
1.0	0	0	$2.3 \cdot 10^{-14}$	$1.9 \cdot 10^{-14}$	$9.3 \cdot 10^{-10}$	$-7.9 \cdot 10^{-7}$	$-8.0 \cdot 10^{-11}$	$-2.1 \cdot 10^{-7}$

Table 38: Case `1a` (`cpv winos`), $r = 9$, Z4: numerator (first lines) and denominator (last lines) coefficients.

8.2 Poles

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$1.11 \cdot 10^{-15} - 1.09 \cdot 10^{-14}\imath$	$5.97 \cdot 10^{-10} - 0.153\imath$	$-1.86 \cdot 10^{-10} + 0.153\imath$
2.0	$-1.11 \cdot 10^{-15} + 4.91\imath$	$-4.13 \cdot 10^{-15} + 0.338\imath$	$-2.87 \cdot 10^{-10} + 0.153\imath$	$1.01 \cdot 10^{-9} - 0.153\imath$
3.0	$-1.11 \cdot 10^{-15} - 4.91\imath$	$-3.5 \cdot 10^{-15} - 0.338\imath$	$1.65 \cdot 10^{-9} - 0.5\imath$	$-1.45 \cdot 10^{-9} + 0.5\imath$
4.0	$-5.55 \cdot 10^{-17} + 1.5\imath$	$-1.64 \cdot 10^{-14} + 0.789\imath$	$-1.26 \cdot 10^{-9} + 0.5\imath$	$2.51 \cdot 10^{-9} - 0.5\imath$
5.0	$-5.55 \cdot 10^{-17} - 1.5\imath$	$-6.96 \cdot 10^{-15} - 0.789\imath$	$3.35 \cdot 10^{-9} - 1.03\imath$	$-3.19 \cdot 10^{-9} + 1.03\imath$
6.0	$-1.39 \cdot 10^{-17} + 0.727\imath$	$-1.08 \cdot 10^{-14} + 1.67\imath$	$-2.7 \cdot 10^{-9} + 1.03\imath$	$5.04 \cdot 10^{-9} - 1.03\imath$
7.0	$-1.39 \cdot 10^{-17} - 0.727\imath$	$-2.77 \cdot 10^{-14} - 1.67\imath$	$7.99 \cdot 10^{-9} - 2.38\imath$	$-6.5 \cdot 10^{-9} + 2.38\imath$
8.0	$1.73 \cdot 10^{-18} + 0.315\imath$	$-5.92 \cdot 10^{-14} + 5.6\imath$	$-6.02 \cdot 10^{-9} + 2.38\imath$	$1.26 \cdot 10^{-8} - 2.38\imath$
9.0	$1.73 \cdot 10^{-18} - 0.315\imath$	$-2.58 \cdot 10^{-14} - 5.6\imath$	$1.63 \cdot 10^{4} - 1.07 \cdot 10^{7}\imath$	$7310.0 + 4.0 \cdot 10^{6}\imath$

Table 39: Case `1a` (`cpv macos`), $r = 9$, Z4: poles.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	0	$-3.67 \cdot 10^{-15} - 3.11 \cdot 10^{-15}\imath$	$-3.27 \cdot 10^{-10} + 2.77 \cdot 10^{-7}\imath$	$2.81 \cdot 10^{-11} + 7.54 \cdot 10^{-8}\imath$
2.0	$-1.11 \cdot 10^{-15} + 4.91\imath$	$-2.8 \cdot 10^{-15} + 0.338\imath$	$1.28 \cdot 10^{-9} - 0.315\imath$	$1.14 \cdot 10^{-10} - 0.315\imath$
3.0	$-1.11 \cdot 10^{-15} - 4.91\imath$	$-5.57 \cdot 10^{-15} - 0.338\imath$	$-2.03 \cdot 10^{-9} + 0.315\imath$	$-4.97 \cdot 10^{-11} + 0.315\imath$
4.0	$-5.55 \cdot 10^{-17} + 1.5\imath$	$3.63 \cdot 10^{-15} + 0.789\imath$	$3.22 \cdot 10^{-9} - 0.727\imath$	$2.35 \cdot 10^{-10} - 0.727\imath$
5.0	$-5.55 \cdot 10^{-17} - 1.5\imath$	$-1.15 \cdot 10^{-14} - 0.789\imath$	$-4.32 \cdot 10^{-9} + 0.727\imath$	$-1.36 \cdot 10^{-10} + 0.727\imath$
6.0	$-1.39 \cdot 10^{-17} + 0.727\imath$	$-2.13 \cdot 10^{-14} - 1.67\imath$	$6.52 \cdot 10^{-9} - 1.5\imath$	$5.02 \cdot 10^{-10} - 1.5\imath$
7.0	$-1.39 \cdot 10^{-17} - 0.727\imath$	$9.08 \cdot 10^{-15} + 1.67\imath$	$-8.83 \cdot 10^{-9} + 1.5\imath$	$-2.53 \cdot 10^{-10} + 1.5\imath$
8.0	$1.73 \cdot 10^{-18} + 0.315\imath$	$-1.42 \cdot 10^{-13} - 5.6\imath$	$1.66 \cdot 10^{-8} - 4.91\imath$	$2.31 \cdot 10^{-9} - 4.91\imath$
9.0	$1.73 \cdot 10^{-18} - 0.315\imath$	$1.23 \cdot 10^{-13} + 5.6\imath$	$-3.32 \cdot 10^{-8} + 4.91\imath$	$-1.3 \cdot 10^{-10} + 4.91\imath$

Table 40: Case `1a` (`cpv winos`), $r = 9$, Z4: poles.

8.3 Zeros

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$1.51 \cdot 10^{-10} + 6.79 \cdot 10^{-8}\imath$	$4.01 \cdot 10^{-10} - 1.92 \cdot 10^{-7}\imath$
2.0	$-8.33 \cdot 10^{-17} + 2.38\imath$	$8.36 \cdot 10^{-16} + 0.163\imath$	$1.08 \cdot 10^{-9} - 0.315\imath$	$-7.87 \cdot 10^{-10} + 0.315\imath$
3.0	$-8.33 \cdot 10^{-17} - 2.38\imath$	$-1.5 \cdot 10^{-15} - 0.163\imath$	$-7.46 \cdot 10^{-10} + 0.315\imath$	$1.69 \cdot 10^{-9} - 0.315\imath$
4.0	$2.78 \cdot 10^{-17} + 1.03\imath$	$-1.15 \cdot 10^{-14} + 0.539\imath$	$2.36 \cdot 10^{-9} - 0.727\imath$	$-2.22 \cdot 10^{-9} + 0.727\imath$
5.0	$2.78 \cdot 10^{-17} - 1.03\imath$	$-3.68 \cdot 10^{-15} - 0.539\imath$	$-1.88 \cdot 10^{-9} + 0.727\imath$	$3.55 \cdot 10^{-9} - 0.727\imath$
6.0	$-6.07 \cdot 10^{-18} + 0.5\imath$	$-1.44 \cdot 10^{-14} + 1.13\imath$	$4.9 \cdot 10^{-9} - 1.5\imath$	$-4.53 \cdot 10^{-9} + 1.5\imath$
7.0	$-6.07 \cdot 10^{-18} - 0.5\imath$	$-1.54 \cdot 10^{-14} - 1.13\imath$	$-3.91 \cdot 10^{-9} + 1.5\imath$	$7.48 \cdot 10^{-9} - 1.5\imath$
8.0	$1.99 \cdot 10^{-17} + 0.153\imath$	$-2.2 \cdot 10^{-14} + 2.68\imath$	$1.81 \cdot 10^{-8} - 4.91\imath$	$-8.23 \cdot 10^{-9} + 4.91\imath$
9.0	$1.99 \cdot 10^{-17} - 0.153\imath$	$-3.49 \cdot 10^{-14} - 2.68\imath$	$-1.09 \cdot 10^{-8} + 4.91\imath$	$3.12 \cdot 10^{-8} - 4.91\imath$

Table 41: Case `1a` (`cpv macos`), $r = 9$, Z4: zeros.

	Opt., $\sigma_9 = 5.1 \cdot 10^{-11}$	LF, $\sigma_9 = 2.7 \cdot 10^{-10}$	AAA-sign-L200, $\sigma_9 = 5.1 \cdot 10^{-11}$	AAA-sign-L1000, $\sigma_9 = 5.1 \cdot 10^{-11}$
1.0	NaN	NaN	$4.67 \cdot 10^{-10} - 0.153\imath$	$6.88 \cdot 10^{-11} - 0.153\imath$
2.0	$1.11 \cdot 10^{-16} + 2.38\imath$	$-3.53 \cdot 10^{-15} + 0.163\imath$	$-1.14 \cdot 10^{-9} + 0.153\imath$	$-1.08 \cdot 10^{-11} + 0.153\imath$
3.0	$1.11 \cdot 10^{-16} - 2.38\imath$	$-4.0 \cdot 10^{-15} - 0.163\imath$	$2.17 \cdot 10^{-9} - 0.5\imath$	$1.67 \cdot 10^{-10} - 0.5\imath$
4.0	$6.94 \cdot 10^{-17} + 1.03\imath$	$2.91 \cdot 10^{-16} + 0.539\imath$	$-3.05 \cdot 10^{-9} + 0.5\imath$	$-9.07 \cdot 10^{-11} + 0.5\imath$
5.0	$6.94 \cdot 10^{-17} - 1.03\imath$	$-8.48 \cdot 10^{-15} - 0.539\imath$	$4.57 \cdot 10^{-9} - 1.03\imath$	$3.34 \cdot 10^{-10} - 1.03\imath$
6.0	$2.08 \cdot 10^{-17} + 0.5\imath$	$4.72 \cdot 10^{-15} + 1.13\imath$	$-6.06 \cdot 10^{-9} + 1.03\imath$	$-1.89 \cdot 10^{-10} + 1.03\imath$
7.0	$2.08 \cdot 10^{-17} - 0.5\imath$	$-1.45 \cdot 10^{-14} - 1.13\imath$	$9.83 \cdot 10^{-9} - 2.38\imath$	$8.7 \cdot 10^{-10} - 2.38\imath$
8.0	$5.2 \cdot 10^{-18} + 0.153\imath$	$-4.7 \cdot 10^{-14} - 2.68\imath$	$-1.44 \cdot 10^{-8} + 2.38\imath$	$-3.2 \cdot 10^{-10} + 2.38\imath$
9.0	$5.2 \cdot 10^{-18} - 0.153\imath$	$3.31 \cdot 10^{-14} + 2.68\imath$	$-2310.0 - 2.66 \cdot 10^{6}\imath$	$4370.0 - 9.93 \cdot 10^{6}\imath$

Table 42: Case `1a` (`cpv winos`), $r = 9$, Z4: zeros.

8.4 Poles and zeros visualisation

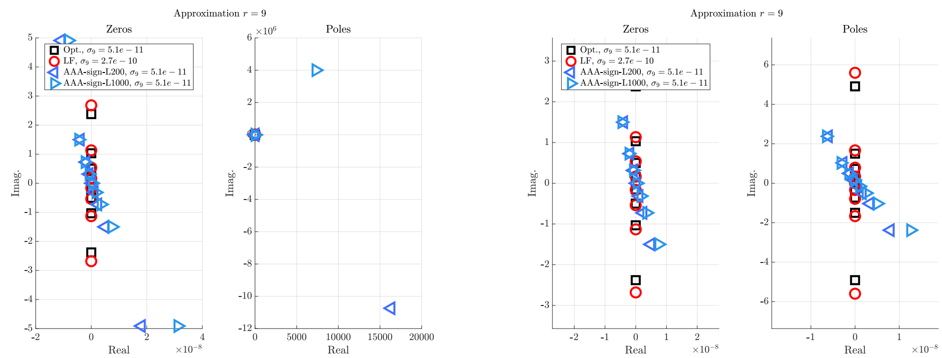

Figure 13: Case '1a', $r = 9$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

8.5 Approximation visualization

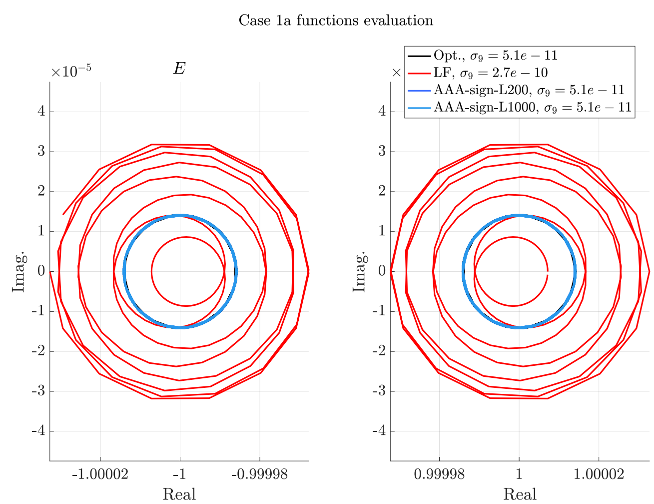

Figure 14: Case '1a', $r = 9$. E and F evaluations.

9 Approximation objective $r = 10$

9.1 Numerator and denominator coefficients

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$	$7.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
z^9	8.7	0	NaN	NaN	8.7	$4.6 \cdot 10^{-3}$	8.7	$4.6 \cdot 10^{-3}$
z^8	0	0	0	0	0.14	0.07	0.14	0.069
z^7	78.0	0	NaN	NaN	78.0	0.13	78.0	0.13
z^6	0	0	0	0	0.036	0.12	0.035	0.12
z^5	120.0	0	NaN	NaN	120.0	0.34	120.0	0.34
z^4	0	0	0	0	-0.31	$-3.8 \cdot 10^{-3}$	-0.31	$-7.4 \cdot 10^{-3}$
z^3	44.0	0	NaN	NaN	44.0	0.17	44.0	0.17
z^2	0	0	0	0	-0.1	-0.016	-0.11	-0.017
z	2.7	0	NaN	NaN	2.8	0.014	2.8	0.014
1.0	0	0	0	0	$-2.7 \cdot 10^{-3}$	$-5.5 \cdot 10^{-4}$	$-2.7 \cdot 10^{-3}$	$-5.6 \cdot 10^{-4}$
z^{10}	1.0	0	NaN	NaN	1.0	$-3.4 \cdot 10^{-17}$	1.0	0
z^9	0	0	0	0	0.051	0.023	0.051	0.022
z^8	34.0	0	NaN	NaN	34.0	0.036	34.0	0.036
z^7	0	0	0	0	0.17	0.12	0.17	0.12
z^6	120.0	0	NaN	NaN	120.0	0.26	120.0	0.26
z^5	0	0	0	0	-0.2	0.062	-0.2	0.057
z^4	89.0	0	NaN	NaN	90.0	0.3	90.0	0.3
z^3	0	0	0	0	-0.24	-0.026	-0.24	-0.028
z^2	14.0	0	NaN	NaN	14.0	0.064	14.0	0.064
z	0	0	0	0	-0.026	$-4.7 \cdot 10^{-3}$	-0.026	$-4.9 \cdot 10^{-3}$
1.0	0.24	0	NaN	NaN	0.24	$1.4 \cdot 10^{-3}$	0.24	$1.4 \cdot 10^{-3}$

Table 43: Case `1a` (`cpv macos`), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$		LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$		AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$		AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$	
Basis	real	imag.	real	imag.	real	imag.	real	imag.
z^{10}	0	0	0	0	$2.4 \cdot 10^{-9}$	$4.3 \cdot 10^{-6}$	$2.2 \cdot 10^{-9}$	$-3.9 \cdot 10^{-8}$
z^9	8.7	0	NaN	NaN	8.7	$-1.0 \cdot 10^{-7}$	8.7	$-1.3 \cdot 10^{-8}$
z^8	0	0	0	0	$5.9 \cdot 10^{-8}$	$1.5 \cdot 10^{-4}$	$7.2 \cdot 10^{-8}$	$-1.4 \cdot 10^{-6}$
z^7	78.0	0	NaN	NaN	78.0	$-2.8 \cdot 10^{-6}$	78.0	$-3.7 \cdot 10^{-7}$
z^6	0	0	0	0	$1.4 \cdot 10^{-7}$	$5.0 \cdot 10^{-4}$	$2.4 \cdot 10^{-7}$	$-5.2 \cdot 10^{-6}$
z^5	120.0	0	NaN	NaN	120.0	$-7.5 \cdot 10^{-6}$	120.0	$-9.7 \cdot 10^{-7}$
z^4	0	0	0	0	$7.2 \cdot 10^{-8}$	$3.8 \cdot 10^{-4}$	$1.8 \cdot 10^{-7}$	$-4.2 \cdot 10^{-6}$
z^3	44.0	0	NaN	NaN	44.0	$-3.7 \cdot 10^{-6}$	44.0	$-4.9 \cdot 10^{-7}$
z^2	0	0	0	0	$7.8 \cdot 10^{-9}$	$6.0 \cdot 10^{-5}$	$2.8 \cdot 10^{-8}$	$-7.1 \cdot 10^{-7}$
z	2.7	0	NaN	NaN	2.7	$-3.0 \cdot 10^{-7}$	2.7	$-3.9 \cdot 10^{-8}$
1.0	0	0	0	0	$9.9 \cdot 10^{-11}$	$1.0 \cdot 10^{-6}$	$4.6 \cdot 10^{-10}$	$-1.3 \cdot 10^{-8}$
z^{10}	1.0	0	NaN	NaN	1.0	0	1.0	0
z^9	0	0	0	0	$1.8 \cdot 10^{-8}$	$3.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-8}$	$-3.5 \cdot 10^{-7}$
z^8	34.0	0	NaN	NaN	34.0	$-8.1 \cdot 10^{-7}$	34.0	$-1.1 \cdot 10^{-7}$
z^7	0	0	0	0	$1.1 \cdot 10^{-7}$	$3.3 \cdot 10^{-4}$	$1.6 \cdot 10^{-7}$	$-3.3 \cdot 10^{-6}$
z^6	120.0	0	NaN	NaN	120.0	$-5.7 \cdot 10^{-6}$	120.0	$-7.4 \cdot 10^{-7}$
z^5	0	0	0	0	$1.2 \cdot 10^{-7}$	$5.2 \cdot 10^{-4}$	$2.5 \cdot 10^{-7}$	$-5.6 \cdot 10^{-6}$
z^4	89.0	0	NaN	NaN	89.0	$-6.5 \cdot 10^{-6}$	89.0	$-8.4 \cdot 10^{-7}$
z^3	0	0	0	0	$2.9 \cdot 10^{-8}$	$1.9 \cdot 10^{-4}$	$8.7 \cdot 10^{-8}$	$-2.1 \cdot 10^{-6}$
z^2	14.0	0	NaN	NaN	14.0	$-1.4 \cdot 10^{-6}$	14.0	$-1.8 \cdot 10^{-7}$
z	0	0	0	0	$1.3 \cdot 10^{-9}$	$1.2 \cdot 10^{-5}$	$5.4 \cdot 10^{-9}$	$-1.4 \cdot 10^{-7}$
1.0	0.24	0	NaN	NaN	0.24	$-2.9 \cdot 10^{-8}$	0.24	$-3.8 \cdot 10^{-9}$

Table 44: Case `1a` (`cpv winos`), $r = 10$, Z4: numerator (first lines) and denominator (last lines) coefficients.

9.2 Poles

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47\imath$	$1.27 \cdot 10^{-14} + 0.141\imath$	$0.00103 - 0.137\imath$	$0.00103 - 0.137\imath$
2.0	$2.22 \cdot 10^{-16} - 5.47\imath$	$1.93 \cdot 10^{-14} - 0.141\imath$	$8.67 \cdot 10^{-4} + 0.138\imath$	$8.68 \cdot 10^{-4} + 0.138\imath$
3.0	$2.78 \cdot 10^{-16} + 1.7\imath$	$1.97 \cdot 10^{-14} + 0.453\imath$	$0.00103 - 0.442\imath$	$0.00103 - 0.442\imath$
4.0	$2.78 \cdot 10^{-16} - 1.7\imath$	$3.14 \cdot 10^{-14} - 0.453\imath$	$5.19 \cdot 10^{-4} + 0.442\imath$	$5.2 \cdot 10^{-4} + 0.442\imath$
5.0	$-1.25 \cdot 10^{-16} + 0.866\imath$	$9.09 \cdot 10^{-16} + 0.891\imath$	$-2.36 \cdot 10^{-4} + 0.867\imath$	$-2.35 \cdot 10^{-4} + 0.867\imath$
6.0	$-1.25 \cdot 10^{-16} - 0.866\imath$	$2.9 \cdot 10^{-14} - 0.891\imath$	$7.39 \cdot 10^{-4} - 0.868\imath$	$7.4 \cdot 10^{-4} - 0.868\imath$
7.0	$-6.94 \cdot 10^{-18} + 0.441\imath$	$-7.64 \cdot 10^{-15} + 1.75\imath$	$-0.00261 + 1.7\imath$	$-0.00261 + 1.7\imath$
8.0	$-6.94 \cdot 10^{-18} - 0.441\imath$	$-1.98 \cdot 10^{-14} - 1.75\imath$	$-7.39 \cdot 10^{-4} - 1.7\imath$	$-7.38 \cdot 10^{-4} - 1.7\imath$
9.0	$-1.34 \cdot 10^{-17} + 0.137\imath$	$-2.66 \cdot 10^{-14} - 5.64\imath$	$-0.0286 + 5.47\imath$	$-0.0285 + 5.47\imath$
10.0	$-1.34 \cdot 10^{-17} - 0.137\imath$	$-5.92 \cdot 10^{-14} + 5.64\imath$	$-0.0228 - 5.49\imath$	$-0.0228 - 5.49\imath$

Table 45: Case `1a` (`cpv macos`), $r = 10$, Z4: poles.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	$2.22 \cdot 10^{-16} + 5.47\imath$	$-1.13 \cdot 10^{-14} + 0.141\imath$	$1.68 \cdot 10^{-9} + 0.137\imath$	$-3.96 \cdot 10^{-10} - 0.137\imath$
2.0	$2.22 \cdot 10^{-16} - 5.47\imath$	$-3.06 \cdot 10^{-14} - 0.141\imath$	$-1.75 \cdot 10^{-9} - 0.137\imath$	$4.93 \cdot 10^{-11} + 0.137\imath$
3.0	$2.78 \cdot 10^{-16} + 1.7\imath$	$6.86 \cdot 10^{-15} + 0.453\imath$	$5.46 \cdot 10^{-9} + 0.441\imath$	$-9.24 \cdot 10^{-10} - 0.441\imath$
4.0	$2.78 \cdot 10^{-16} - 1.7\imath$	$-3.6 \cdot 10^{-14} - 0.453\imath$	$-5.53 \cdot 10^{-9} - 0.441\imath$	$5.0 \cdot 10^{-10} + 0.441\imath$
5.0	$-1.25 \cdot 10^{-16} + 0.866\imath$	$1.56 \cdot 10^{-14} + 0.891\imath$	$1.05 \cdot 10^{-8} + 0.866\imath$	$-1.72 \cdot 10^{-9} - 0.866\imath$
6.0	$-1.25 \cdot 10^{-16} - 0.866\imath$	$-4.39 \cdot 10^{-14} - 0.891\imath$	$-1.08 \cdot 10^{-8} - 0.866\imath$	$1.04 \cdot 10^{-9} + 0.866\imath$
7.0	$-6.94 \cdot 10^{-18} + 0.441\imath$	$8.07 \cdot 10^{-15} + 1.75\imath$	$2.0 \cdot 10^{-8} + 1.7\imath$	$-3.56 \cdot 10^{-9} - 1.7\imath$
8.0	$-6.94 \cdot 10^{-18} - 0.441\imath$	$-5.94 \cdot 10^{-14} - 1.75\imath$	$-2.13 \cdot 10^{-8} - 1.7\imath$	$1.79 \cdot 10^{-9} + 1.7\imath$
9.0	$-1.34 \cdot 10^{-17} + 0.137\imath$	$-1.74 \cdot 10^{-13} - 5.64\imath$	$5.76 \cdot 10^{-8} + 5.47\imath$	$-1.63 \cdot 10^{-8} - 5.47\imath$
10.0	$-1.34 \cdot 10^{-17} - 0.137\imath$	$1.08 \cdot 10^{-13} + 5.64\imath$	$-7.4 \cdot 10^{-8} - 5.47\imath$	$7.27 \cdot 10^{-10} + 5.47\imath$

Table 46: Case `1a` (`cpv winos`), $r = 10$, Z4: poles.

9.3 Zeros

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 4.3 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 4.3 \cdot 10^{-12}$
1.0	NaN	NaN	$9.66 \cdot 10^{-4} + 1.92 \cdot 10^{-4}\imath$	$9.67 \cdot 10^{-4} + 1.96 \cdot 10^{-4}\imath$
2.0	0	$1.35 \cdot 10^{-14} - 8.67 \cdot 10^{-15}\imath$	$0.00105 - 0.282\imath$	$0.00105 - 0.282\imath$
3.0	$3.33 \cdot 10^{-16} + 2.67\imath$	$1.74 \cdot 10^{-14} + 0.289\imath$	$7.24 \cdot 10^{-4} + 0.282\imath$	$7.24 \cdot 10^{-4} + 0.282\imath$
4.0	$3.33 \cdot 10^{-16} - 2.67\imath$	$2.58 \cdot 10^{-14} - 0.289\imath$	$9.39 \cdot 10^{-4} - 0.63\imath$	$9.39 \cdot 10^{-4} - 0.63\imath$
5.0	$2.43 \cdot 10^{-16} + 1.19\imath$	$1.34 \cdot 10^{-14} + 0.647\imath$	$2.23 \cdot 10^{-4} + 0.63\imath$	$2.23 \cdot 10^{-4} + 0.63\imath$
6.0	$2.43 \cdot 10^{-16} - 1.19\imath$	$3.58 \cdot 10^{-14} - 0.647\imath$	$-0.00102 + 1.19\imath$	$-0.00102 + 1.19\imath$
7.0	$4.16 \cdot 10^{-17} + 0.629\imath$	$-5.1 \cdot 10^{-15} + 1.23\imath$	$3.05 \cdot 10^{-4} - 1.19\imath$	$3.06 \cdot 10^{-4} - 1.19\imath$
8.0	$4.16 \cdot 10^{-17} - 0.629\imath$	$6.08 \cdot 10^{-15} - 1.23\imath$	$-0.00688 + 2.67\imath$	$-0.00687 + 2.67\imath$
9.0	$-3.47 \cdot 10^{-18} + 0.281\imath$	$-2.24 \cdot 10^{-14} + 2.75\imath$	$-0.00398 - 2.67\imath$	$-0.00398 - 2.67\imath$
10.0	$-3.47 \cdot 10^{-18} - 0.281\imath$	$-3.62 \cdot 10^{-14} - 2.75\imath$	$-971.0 + 392.0\imath$	$-975.0 + 389.0\imath$

Table 47: Case `1a` (`cpv macos`), $r = 10$, Z4: zeros.

	Opt., $\sigma_{10} = 3.6 \cdot 10^{-12}$	LF, $\sigma_{10} = 8.4 \cdot 10^{-12}$	AAA-sign-L200, $\sigma_{10} = 3.6 \cdot 10^{-12}$	AAA-sign-L1000, $\sigma_{10} = 3.6 \cdot 10^{-12}$
1.0	NaN	NaN	$-3.62 \cdot 10^{-11} - 3.64 \cdot 10^{-7}\imath$	$-1.7 \cdot 10^{-10} + 4.59 \cdot 10^{-9}\imath$
2.0	0	$-2.24 \cdot 10^{-14} - 5.81 \cdot 10^{-15}\imath$	$3.48 \cdot 10^{-9} + 0.281\imath$	$-6.42 \cdot 10^{-10} - 0.281\imath$
3.0	$3.33 \cdot 10^{-16} + 2.67\imath$	$-4.04 \cdot 10^{-16} + 0.289\imath$	$-3.55 \cdot 10^{-9} - 0.281\imath$	$2.69 \cdot 10^{-10} + 0.281\imath$
4.0	$3.33 \cdot 10^{-16} - 2.67\imath$	$-3.45 \cdot 10^{-14} - 0.289\imath$	$7.74 \cdot 10^{-9} + 0.629\imath$	$-1.27 \cdot 10^{-9} - 0.629\imath$
5.0	$2.43 \cdot 10^{-16} + 1.19\imath$	$-3.9 \cdot 10^{-14} - 0.647\imath$	$-7.85 \cdot 10^{-9} - 0.629\imath$	$7.51 \cdot 10^{-10} + 0.629\imath$
6.0	$2.43 \cdot 10^{-16} - 1.19\imath$	$1.22 \cdot 10^{-14} + 0.647\imath$	$1.43 \cdot 10^{-8} + 1.19\imath$	$-2.4 \cdot 10^{-9} - 1.19\imath$
7.0	$4.16 \cdot 10^{-17} + 0.629\imath$	$1.25 \cdot 10^{-14} + 1.23\imath$	$-1.48 \cdot 10^{-8} - 1.19\imath$	$1.38 \cdot 10^{-9} + 1.19\imath$
8.0	$4.16 \cdot 10^{-17} - 0.629\imath$	$-5.01 \cdot 10^{-14} - 1.23\imath$	$3.04 \cdot 10^{-8} + 2.67\imath$	$-6.13 \cdot 10^{-9} - 2.67\imath$
9.0	$-3.47 \cdot 10^{-18} + 0.281\imath$	$-8.32 \cdot 10^{-14} - 2.75\imath$	$-3.4 \cdot 10^{-8} - 2.67\imath$	$2.21 \cdot 10^{-9} + 2.67\imath$
10.0	$-3.47 \cdot 10^{-18} - 0.281\imath$	$2.31 \cdot 10^{-14} + 2.75\imath$	$-1130.0 + 2.0 \cdot 10^{6}\imath$	$-1.28 \cdot 10^{7} - 2.24 \cdot 10^{8}\imath$

Table 48: Case `1a` (`cpv winos`), $r = 10$, Z4: zeros.

9.4 Poles and zeros visualisation

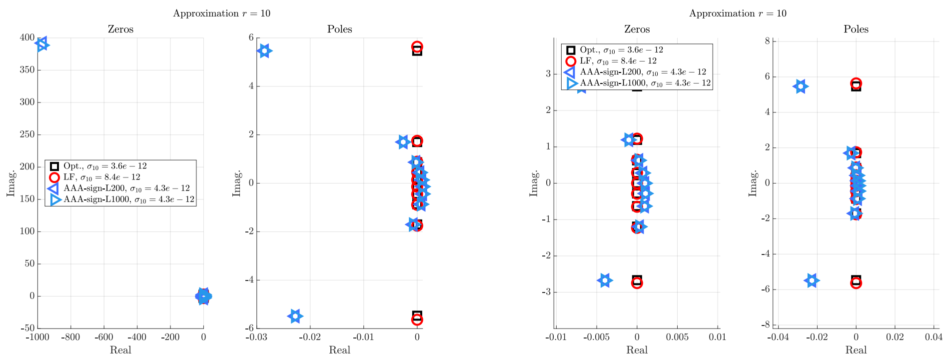

Figure 15: Case `'1a'`, $r = 10$. Left frame: zeros (left) and poles (right). Right frame: same but discarding the largest real entry.

9.5 Approximation visualization

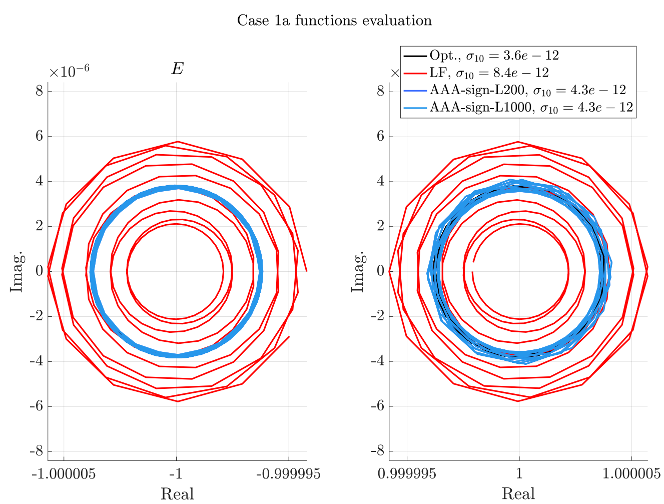

Figure 16: Case `'1a'`, $r = 10$. E and F evaluations.